nature communications

Article

https://doi.org/10.1038/s41467-026-77349-8

Microenvironment-aware transcriptome reconstruction in spatial transcriptomics

Received: 31 October 2025

Accepted: 20 August 2026

Published online: 03 September 2026

Check for updates

Shi-Tong Yang[1], Pai Peng[2], Hui-Feng He[1], Meng-Guo Wang[1], Bo-Han Si[1], Xiao-Fei Zhang[1,3,4] ✉ & Luonan Chen[5,6] ✉

Imaging-based spatial transcriptomics provides single-cell resolution, but remains limited to targeted gene panels, leaving much of the transcriptome and cellular-state variation unobserved. Existing approaches infer unmeasured genes primarily through transcriptional alignment between spatial and reference datasets, which effectively recover broad cell identities, but may not fully capture context-dependent variation within cell types. Here we show that local microenvironmental organization is associated with transcriptional variation beyond cell identity and provides complementary constraints for transcriptome reconstruction. Based on this observation, we present Emerge, a microenvironment-aware framework that integrates transcriptional identity with the local microenvironment to reconstruct transcriptome-scale expression. Across MERFISH and Xenium datasets spanning neural and tumor ecosystems, we show that Emerge improves recovery of structured transcriptional variation and reveals coordinated relationships among cellular states, ecological niches and intercellular communication programs that are only partially captured by targeted gene panels. Together, we demonstrate that Emerge extends transcriptome reconstruction from gene completion to context-aware recovery of cellular states.

Cell identity alone does not fully capture cellular state. Cells sharing the same identity can adopt distinct transcriptional programs depending on the tissue environments[1,2]. Characterizing this context-dependent variation is central to understanding tissue organization, cellular plasticity, and disease progression. Recent imaging-based spatial transcriptomics technologies, including MERFISH[3], Xenium[4], and CosMx[5], enable transcriptomic profiling at single-cell resolution while preserving spatial context[3,4,6]. These technologies have transformed the study of cellular organization in intact tissues and enabled systematic investigation of how cellular states vary across spatially structured microenvironments. However, because current imaging-based platforms typically measure only hundreds to a few thousand genes, they often provide sufficient information to identify major cell types but remain limited in their ability to resolve fine-grained cellular states and context-dependent transcriptional programs[7–9].

To overcome this limitation, numerous computational approaches have been developed to infer unmeasured genes by leveraging reference single-cell or single-nucleus RNA sequencing (sc/snRNA-seq) datasets[10–12]. Existing methods for transcriptome completion can be broadly categorized into two classes. The first aligns spatial cells with transcriptionally similar reference cells and transfers expression information through nearest-neighbor matching, latent-variable modeling, or optimal transport[12–21]. The second incorporates spatial information by encouraging nearby cells to exhibit similar expression

[1]School of Mathematics and Statistics, and Hubei Key Laboratory of Mathematical Sciences, Central China Normal University, Wuhan, China. [2]School of Artificial Intelligence, Jianghan University, Wuhan, China. [3]Key Laboratory of Nonlinear Analysis & Applications (Ministry of Education), Central China Normal University, Wuhan, China. [4]Center for Intelligent Drug Discovery, Academy of Frontier Interdisciplinary Research, Central China Normal University, Wuhan, China. [5]School of Mathematical Sciences and School of AI, Shanghai Jiao Tong University, Shanghai, China. [6]Key Laboratory of Systems Health Science of Zhejiang Province, School of Life Science, Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. ✉e-mail: zhangxf@ccnu.edu.cn; lnchen@sjtu.edu.cn

patterns or by smoothing predictions across local neighborhoods[22–25]. These approaches have substantially expanded the utility of panel-limited spatial transcriptomics datasets and improved recovery of missing genes. Nevertheless, they are largely built upon the assumption that transcriptional similarity and spatial proximity are sufficient to recover unmeasured expression.

Increasing evidence suggests that this assumption is incomplete. Cellular states are shaped not only by intrinsic transcriptional identity but also by interactions with local microenvironments[2,26–28]. Cells of the same type can adopt distinct functional programs when embedded within different ecological niches, and many biologically important transcriptional programs are closely linked to local cellular composition, tissue architecture and intercellular communication[29–32]. Consistent with this view, recent studies have shown that neighborhood composition frequently explains substantial transcriptional variation beyond cell identity alone[33–39]. These observations suggest that the local microenvironmental context contains biologically meaningful information that may help constrain transcriptome reconstruction. However, current reconstruction frameworks do not explicitly model how microenvironmental organization shapes transcriptional variation within cell types.

Here we present Emerge, a microenvironment-aware framework for transcriptome-scale reconstruction of imaging-based spatial transcriptomics. Motivated by the observation that cellular states are jointly shaped by intrinsic identity and local microenvironmental context, Emerge integrates transcriptomic similarity with niche-associated information to guide transcriptome reconstruction. By incorporating local microenvironmental organization as a complementary biological constraint, Emerge extends transcriptome reconstruction from recovering missing genes to recovering context-dependent transcriptional variation.

Across neural and tumor ecosystems profiled by MERFISH and Xenium, Emerge consistently improved transcriptome reconstruction and enabled recovery of biologically meaningful cellular states that were only partially captured by measured genes alone. Reconstructed transcriptomes revealed coordinated relationships among cellular states, ecological niches, and intercellular communication programs across multiple tissues and disease contexts, supporting microenvironment-aware reconstruction as a general strategy for recovering context-dependent cellular organization beyond targeted gene panels.

Results

Overview of Emerge

Emerge is a computational framework that extends imaging-based spatial transcriptomics (ST) from targeted panels to transcriptome-scale, microenvironment-aware expression maps. Although imaging-based ST provides cellular-resolution measurements, limited gene panels can underrepresent fine-grained cellular states and intercellular communication programs (Supplementary Fig. 1). Emerge addresses this limitation by integrating spatial measurements with biologically compatible sc/snRNA-seq reference to reconstruct unmeasured genes while preserving both cell identity and niche-associated transcriptional variation (Fig. 1a). The framework is based on the premise that cell identity defines broad differences across cell types, whereas local microenvironments shape continuous variation within them (Supplementary Note 1).

Emerge comprises two core inference modules and an integrated downstream analysis framework. The Cell Type Annotator establishes an identity backbone by assigning spatial cells to reference-defined cell types using an ensemble strategy that produces probabilistic labels and confidence scores (Fig. 1b). High-confidence cells serve as anchors for downstream reconstruction, whereas low-confidence cells are retained as uncertain rather than forced into potentially incorrect identities. This confidence-aware design introduces type constraints as an identity-preserving inductive bias that reduces biologically implausible cross-type alignments under limited gene coverage.

The Gene Expression Predictor reconstructs transcriptomes using a type-constrained fused Gromov-Wasserstein optimal transport (FGW-OT) framework[40] (Fig. 1c). The model integrates complementary constraints derived from cell identity, measured gene expression, and local microenvironmental organization. Shared-gene similarity establishes transcriptionally coherent alignment between spatial and reference cells, whereas microenvironmental relational structure refines alignment according to local tissue context. By combining identity constraints, expression similarity, and microenvironment-aware regularization within a unified framework, Emerge recovers context-dependent transcriptional variation while remaining robust to incomplete reference support. Low-confidence cells are subsequently reconstructed through neighborhood-guided regression from nearby high-confidence cells, enabling uncertainty-aware transcriptome recovery (Fig. 1d).

Building on reconstructed transcriptomes and environmental descriptors, the Microenvironment-aware Spatial Analyzer links cellular states to local niche composition, identifies spatially organized tissue regions, and integrates state-niche relationships across sections or samples (Fig. 1e). This framework enables transcriptome-scale characterization of microenvironment-dependent cellular organization in intact tissues.

Microenvironment-aware reconstruction improves recovery of within-type transcriptional variation in mouse MOp

We first applied Emerge to MERFISH (Multiplexed Error-Robust Fluorescence In Situ Hybridization) data from the mouse primary motor cortex (MOp)[41] together with matched single-nucleus RNA-seq references[42] to investigate whether local microenvironmental organization provides information complementary to transcriptomic similarity for transcriptome reconstruction. Environmental descriptors were constructed from neighboring cell-type composition and excluded the identity of the query cell itself. Environmental composition exhibited substantially stronger associations with gene expression than spatial coordinates alone, both within and across cell types (Fig. 2a), suggesting that local niche organization captures biologically relevant information beyond spatial position and may therefore provide complementary signals for transcriptome reconstruction.

We next evaluated transcriptome reconstruction performance using five-fold cross-validation on MERFISH-measured genes. Emerge_cell and its cluster-level variant (Emerge_cluster) were benchmarked against eight representative methods spanning shared-gene alignment, transfer learning, ensemble learning, and geometry-aware optimal transport approaches. Across both gene- and cell-level evaluations, Emerge consistently ranked among the top-performing methods across multiple metrics (Supplementary Note 2.1), with aggregated ranks confirming robust performance across complementary evaluation criteria (Fig. 2b, c and Supplementary Fig. 2). Comparable performance between Emerge_cell and Emerge_cluster further indicated that the framework remains effective under different annotation modes.

To directly assess the contribution of microenvironment-aware modeling, we performed ablation and sensitivity analyses of the balance parameter α. Reconstruction accuracy exhibited a clear unimodal relationship with α, with the full model ($\alpha = 0.5$) improving accuracy by 3.95% relative to transcriptomic alignment alone ($\alpha = 0$; two-sided paired Wilcoxon signed-rank test, $p = 3.5 \times 10^{-9}$; Supplementary Fig. 3a, b). Environmental alignment alone ($\alpha = 1$) also underperformed the integrated model, indicating that accurate reconstruction requires both transcriptomic and microenvironmental information. In addition, both Emerge_cell and Emerge_cluster outperformed their first-order-only counterparts (Supplementary Fig. 3c). The optimal α further shifted systematically with measured gene panel size (Supplementary Fig. 4), with lower-coverage panels placing greater weight on

https://doi.org/10.1038/s41467-026-77349-8

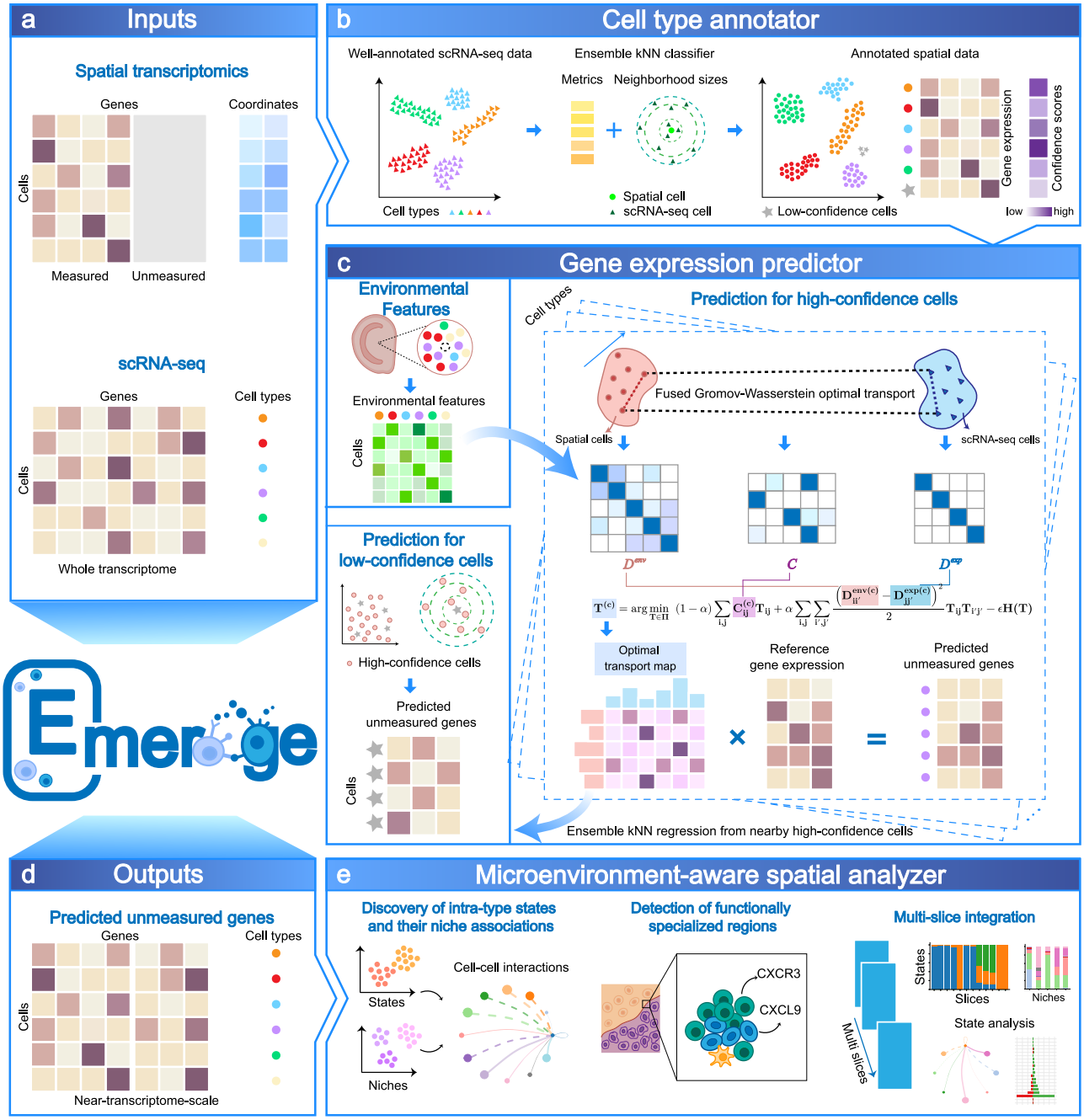


Fig. 1 | Overview of Emerge. a Emerge integrates imaging-based spatial transcriptomics with biologically compatible sc/snRNA-seq reference datasets to reconstruct transcriptome-scale expression. Inputs include measured gene expression, cell coordinates, and annotated reference transcriptomes. **b** Cell type annotator. Spatial cells are assigned to reference-defined cell types using an ensemble annotation strategy that produces probabilistic labels and confidence scores. High-confidence cells provide a reliable identity backbone for downstream reconstruction, whereas low-confidence cells are handled separately to reduce propagation of uncertain annotations. **c** Gene expression predictor. For high-confidence cells, unmeasured genes are inferred through type-constrained alignment between spatial and reference cells by jointly leveraging measured-gene similarity and local microenvironmental organization. For low-confidence cells, expression is estimated from nearby high-confidence cells through neighborhood-guided regression. **d** Emerge generates transcriptome-scale expression profiles together with cell-type annotations, enabling reconstruction of single-cell spatial transcriptomes in their native tissue context. **e** Microenvironment-aware spatial analyzer. Reconstructed transcriptomes support downstream analyses of cellular states, ecological niches, intercellular communication, and conserved spatial programs across tissue sections.

microenvironmental information and higher-coverage panels relying increasingly on transcriptomic similarity, indicating that the relative contribution of the two information sources depends on panel coverage (Supplementary Note 2.2).

To evaluate robustness, we further assessed Emerge across alternative annotation strategies[43,44] (Supplementary Note 2.3), microenvironmental feature definitions[45,46] (Supplementary Note 2.4) and independent references derived from different biological cohorts and sequencing technologies(Supplementary Note 2.5). Consistent performance was observed in all settings (Supplementary Figs. 5–7), supporting the robustness of Emerge to realistic analytical and biological variation.

https://doi.org/10.1038/s41467-026-77349-8

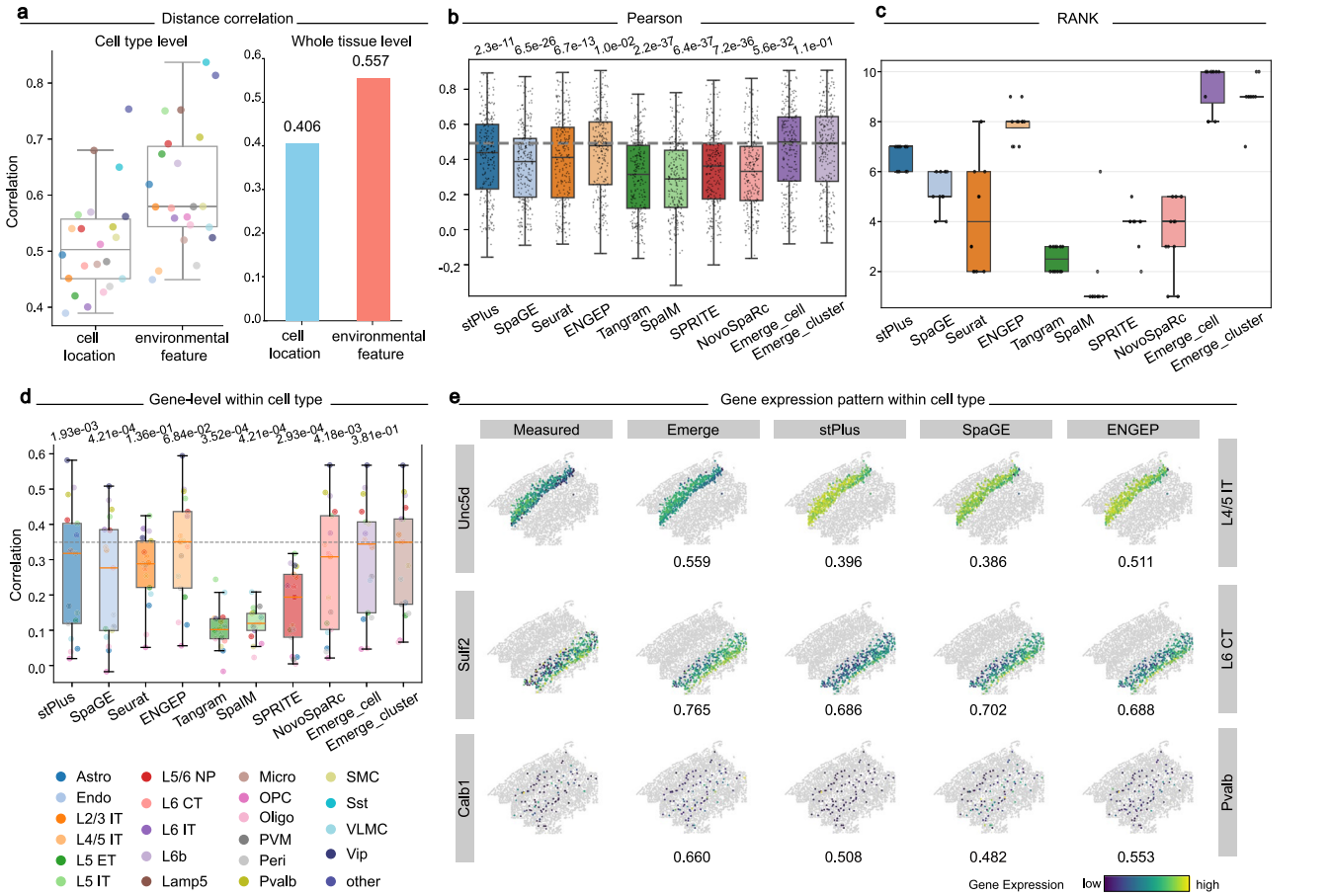


Fig. 2 | Microenvironment-aware reconstruction captures structured transcriptional variation in the mouse primary motor cortex (MOp). a Distance correlations between gene expression and either cell location or environmental features. Correlations are calculated across individual cell types ($n = 22$ cell types; left, each point represents one cell type) and across all cells in the tissue (right). Numbers above bars indicate the corresponding distance correlation values. **b** Gene-level reconstruction performance measured by Pearson correlation between predicted and measured expression across genes. Each box summarizes the distribution of gene-wise Pearson correlations ($n = 249$ genes) for each method. Box plots show median values, interquartile ranges, and whiskers extending to 1.5 × IQR. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests, with exact p values shown above the box plots. **c** AR (aggregated accuracy rank) scores integrating gene-level and cell-level reconstruction performance across six evaluation metrics. Higher AR scores indicate better overall performance. **d** Within-cell-type reconstruction performance evaluated using structured genes. Pearson correlations between predicted and measured expression are shown for each method. Each dot represents one cell type ($n = 22$ cell types), with values corresponding to the mean gene-wise Pearson correlation across structured genes. Colors denote different cell types. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests. **e** Spatial expression patterns of representative genes (*Unc5d*, *Sulf2* and *Calb1*) within selected neuronal cell types (L2/3 IT, L5 ET and L6 CT). Measured expression and reconstructed expression from different methods are shown side by side. Numbers indicate Pearson correlations between predicted and measured expression. Source data are provided as a Source Data file.

Although overall accuracy evaluates transcriptome-scale recovery, the primary objective of Emerge is to recover biologically meaningful variation within cell types. We therefore focused on structured genes exhibiting substantial within-cell-type variability and spatial organization. Across these genes, Emerge consistently outperformed most competing approaches (Fig. 2d). Similar trends were observed for highly variable genes (HVGs), spatially variable genes (SVGs) and marker genes, whereas performance differences were substantially reduced for non-HVG and non-SVG control genes (Supplementary Figs. 8 and 9). These results indicate that the benefits of Emerge are concentrated in biologically informative genes exhibiting pronounced transcriptional or spatial structure, precisely the types of signals expected to benefit from incorporating local microenvironmental context (Supplementary Note 2.6).

Representative examples further illustrated these improvements. At the tissue scale, Emerge faithfully reconstructed major spatial expression programs associated with distinct neuronal populations, preserving the large-scale organization of layer- and cell-type-specific expression patterns (Supplementary Fig. 10). More importantly, at the within-cell-type level, Emerge more accurately recovered spatially structured transcriptional variation, including the layer- and cell-type-restricted expression patterns of *Unc5d*, *Sulf2* and *Calb1* (Fig. 2e). Similar improvements were observed across additional neuronal marker genes (Supplementary Fig. 11), supporting the ability of microenvironment-aware reconstruction to recover biologically meaningful variation within cell types.

Together, these analyses indicate that local microenvironmental context provides information complementary to transcriptomic similarity and is particularly informative for recovering structured transcriptional variation within cell types. By integrating both sources of information, Emerge improves reconstruction of biologically meaningful spatial gene-expression programs across multiple cellular scales.

Emerge reveals astrocyte state-niche organization in mouse MOp

We next asked whether transcriptome-scale reconstruction could expand the biological information accessible from imaging-based spatial transcriptomics beyond that provided by targeted gene panels alone. Applying Emerge-generated anatomically coherent expression

patterns for unmeasured genes that closely matched independent in situ hybridization (ISH) data from the Allen Brain Atlas[47] (Fig. 3a). Broad neuronal markers such as *Snap25* and spatially restricted genes, including *Tac1* and *Adarb2*[42] exhibited expression patterns consistent with known cortical organization, supporting the biological validity of the reconstructed transcriptomes.

Integrating reconstructed genes with measured transcripts yielded clearer separation of major cellular lineages and improved organization of cellular states relative to competing approaches (Fig. 3b and Supplementary Fig. 12a). Furthermore, transcriptome-scale profiles consistently resolved additional cellular states across cell types and clustering resolutions compared with the measured panel alone (Supplementary Fig. 12b). These observations suggest that transcriptome reconstruction expands the effective transcriptional feature space available for downstream analysis. We therefore asked whether reconstructed transcriptomes could reveal cellular states associated with local microenvironmental organization.

Focusing on astrocytes, whose transcriptional diversity is strongly influenced by local tissue context[48], we identified two major transcriptional states with distinct spatial distributions (Fig. 3c). Environmental descriptors further partitioned astrocytes into four niche types characterized by distinct cellular compositions (Fig. 3d, e and Supplementary Fig. 13b). State-niche enrichment analysis revealed a strong correspondence between astrocyte states and local environments, with State 1 enriched in Niches 1–3 and depleted from Niche 4, whereas State 2 exhibited the opposite pattern (Fig. 3f and Supplementary Fig. 13c, d). The two states also displayed distinct functional programs (Fig. 3g and Supplementary Fig. 13e, f and Supplementary Data 1). State 1 was enriched for developmental and remodeling-associated processes, including morphogenesis, cell proliferation, and protein turnover, whereas State 2 showed strong enrichment for oxidative and lipid metabolic pathways, including fatty-acid β-oxidation, cellular respiration, and energy-generation processes. Together, these findings suggest that astrocyte state variation in the MOp is closely coupled to local niche organization and associated functional programs[49,50].

Cell-cell communication analysis provided further evidence that the two astrocyte states occupied distinct signaling environments (Fig. 3h, j and Supplementary Fig. 13g, h). When astrocytes were considered as receiver populations, State 1 preferentially received inputs from excitatory neuronal populations, whereas State 2 received stronger inputs from deep-layer neuronal, vascular and glial-associated populations (Fig. 3h). Consistent with these differences, State 1 exhibited higher activity of neuronal adhesion and synaptic-interaction receptors, whereas State 2 preferentially utilized growth-factor- and morphogen-related receptors[49,51] (Fig. 3i). At the pathway level, State 1 was associated with extracellular matrix organization, receptor tyrosine kinase signaling and integrin cell-surface interactions, whereas State 2 showed stronger GPCR-, G-protein- and chemokine-receptor signaling activity (Fig. 3j). Together, these findings suggest that distinct astrocyte states are embedded within specialized microenvironmental niches characterized by coordinated differences in cellular composition, functional programs and intercellular communication.

We next examined whether these niche-associated astrocyte states could also be recovered using transcriptomes reconstructed by alternative methods. Although several competing approaches produced heterogeneous astrocyte assignments, most failed to recover the coordinated relationship between astrocyte states and local niches, either collapsing cells into a dominant state or generating diffuse state assignments with weak niche correspondence (Fig. 3k and Supplementary Fig. 14). In contrast, Emerge recovered astrocyte states that were both transcriptionally distinct and specifically associated with local environmental niches.

Together, these findings indicate that coordinated state-niche-communication organization can be recovered from reconstructed transcriptomes and is largely missed by existing reconstruction approaches. By integrating transcriptomic similarity with microenvironment-associated relational information, Emerge reveals how local tissue context shapes cellular states and their interactions within intact spatial ecosystems.

Microenvironment-aware reconstruction improves recovery of structured transcriptional variation in breast cancer

We next evaluated Emerge in a human breast cancer Xenium dataset integrated with matched scRNA-seq references[4] to determine whether microenvironmental information becomes increasingly important in complex pathological tissues. Unlike the layered architecture of the mouse cortex, breast tumors comprise highly heterogeneous ecosystems in which malignant, stromal, endothelial, and immune populations are extensively intermixed. In this setting, environmental composition exhibited substantially stronger associations with gene expression than spatial coordinates alone, both within individual cell types and across the entire tissue (Fig. 4a), suggesting that local cellular organization provides a more informative representation of regulatory context than physical location.

To quantify the contribution of microenvironmental information, we performed ablation analyses of the balance parameter α. Reconstruction accuracy exhibited a clear unimodal relationship with α, with the full model (α = 0.5) improving accuracy by 19.16% relative to transcriptomic alignment alone (α = 0; two-sided paired Wilcoxon signed-rank test, $p = 8.9 \times 10^{-52}$; Supplementary Fig. 15a, b). Environmental alignment alone (α = 1) also underperformed the integrated model, indicating that transcriptomic similarity and microenvironmental context provide complementary constraints. Consistently, both Emerge_cell and Emerge_cluster outperformed their first-order-only counterparts (Supplementary Fig. 15c). Notably, the gain from incorporating microenvironmental information was substantially larger than that observed in the mouse cortex, suggesting that local cellular interactions become increasingly important determinants of transcriptional variation in tumor ecosystems.

We next benchmarked Emerge against representative reconstruction methods using five-fold cross-validation. Across both gene- and cell-level evaluations, Emerge consistently ranked among the top-performing approaches across multiple metrics (Fig. 4b, c and Supplementary Fig. 16). Performance remained stable across alternative annotation strategies, environmental feature definitions, and independent scRNA-seq references derived from distinct breast cancer cohorts[52] (Supplementary Figs. 17–19), indicating robust performance across analytical settings.

Having established the contribution of microenvironmental information at the global level, we next examined whether these gains extend to biologically meaningful within-cell-type transcriptional variation. Across structured genes exhibiting both substantial within-cell-type variability and spatial organization, Emerge consistently achieved the largest improvements relative to competing approaches (Fig. 4d and Supplementary Fig. 20). Similar trends were observed for HVGs, SVGs, and marker genes, whereas performance differences were substantially reduced for non-HVG and non-SVG control genes (Supplementary Fig. 21). These findings indicate that microenvironment-aware modeling preferentially improves recovery of structured transcriptional variation associated with cellular states and local tissue context.

Representative examples further illustrated these advantages. At the tissue scale, Emerge accurately recovered spatial expression programs associated with tumor, stromal, endothelial and DCIS-associated (ductal carcinoma in situ-associated) compartments (Supplementary Fig. 22). More importantly, Emerge preserved fine-grained spatial heterogeneity within individual cell types, including localized expression domains in myoepithelial, endothelial and macrophage populations as well as stromal regions surrounding DCIS lesions (Fig. 4e and Supplementary Fig. 23). Across these examples, Emerge

https://doi.org/10.1038/s41467-026-77349-8

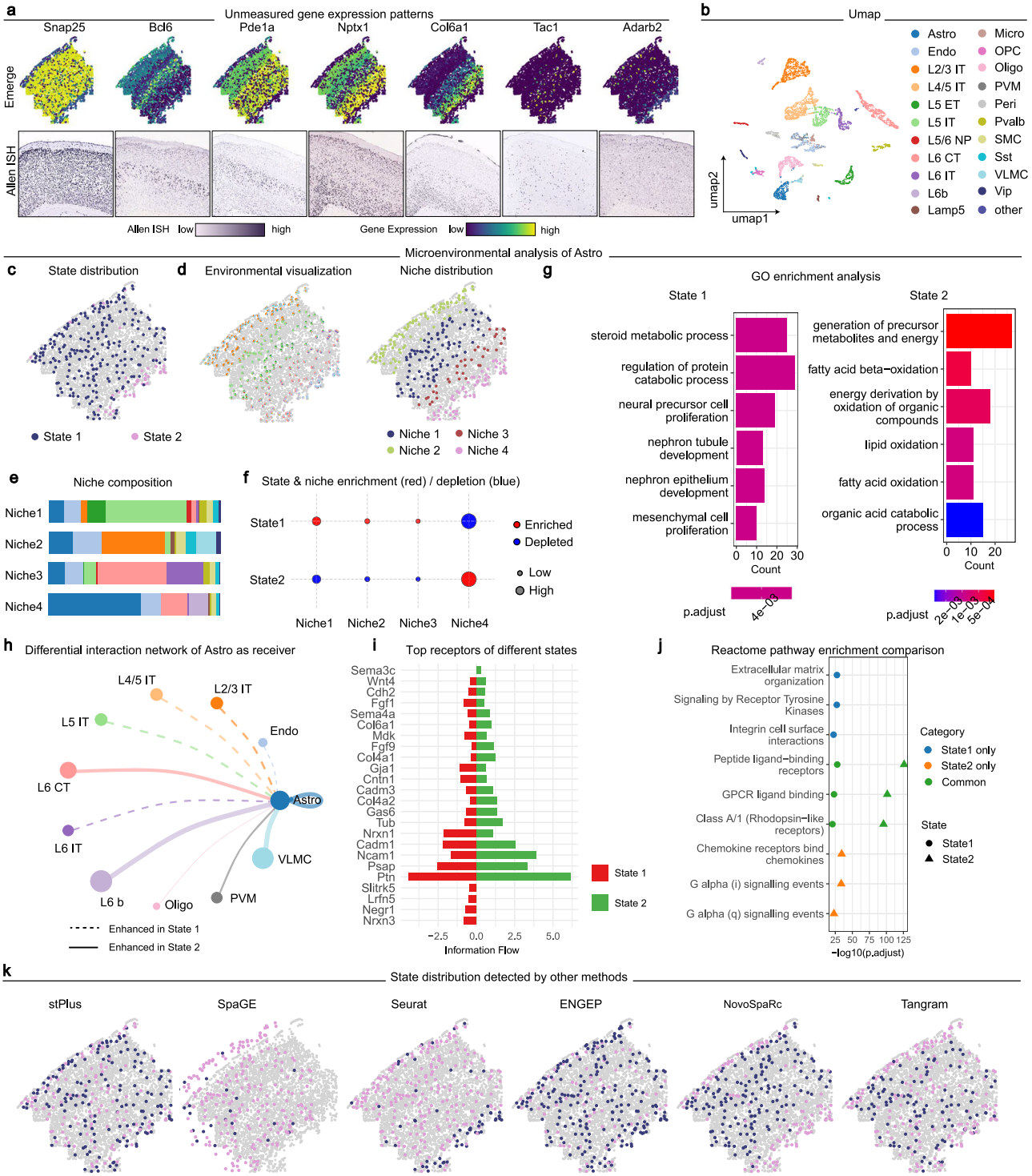


Fig. 3 | Transcriptome-scale reconstruction reveals niche-associated astrocyte states in mouse MOp. a Reconstructed spatial expression patterns of representative unmeasured genes (top row) compared with corresponding in situ hybridization (ISH) images (bottom row) from the Allen Brain Atlas. **b** UMAP (Uniform Manifold Approximation and Projection) embedding of spatial cells based on combined measured and reconstructed gene expression. **c**–**j** Microenvironment-aware analysis of astrocytes. **c** Visualization of two astrocyte states. **d** Visualization of environmental features (left) and hierarchical clustering of environmental profiles defining four microenvironmental niches (right). **e** Cell-type composition of each niche. **f** Enrichment (red dots) and depletion (blue dots) of cell states across niches assessed by hypergeometric tests with Benjamini-Hochberg FDR correction. Dot size represents $-\log_{10}$(adjusted p value). **g** Gene Ontology enrichment of marker genes for astrocyte States 1 and 2. Enrichment was assessed by clusterProfiler-based over-representation analysis with Benjamini-Hochberg correction. **h** Differential ligand-receptor interaction network for astrocytes as receivers inferred using CellChat. Dashed and solid lines denote interactions preferentially enriched in States 1 and 2, respectively. **i** Top predicted receptor genes for States 1 and 2 with state-specific differences in ligand-receptor interaction strength. **j** Reactome pathway enrichment analysis of state-specific ligands and receptors. Enrichment significance was assessed by over-representation analysis with Benjamini-Hochberg adjusted p values. **k** Spatial distributions of astrocyte states inferred by alternative methods. Source data are provided as a Source Data file.

https://doi.org/10.1038/s41467-026-77349-8

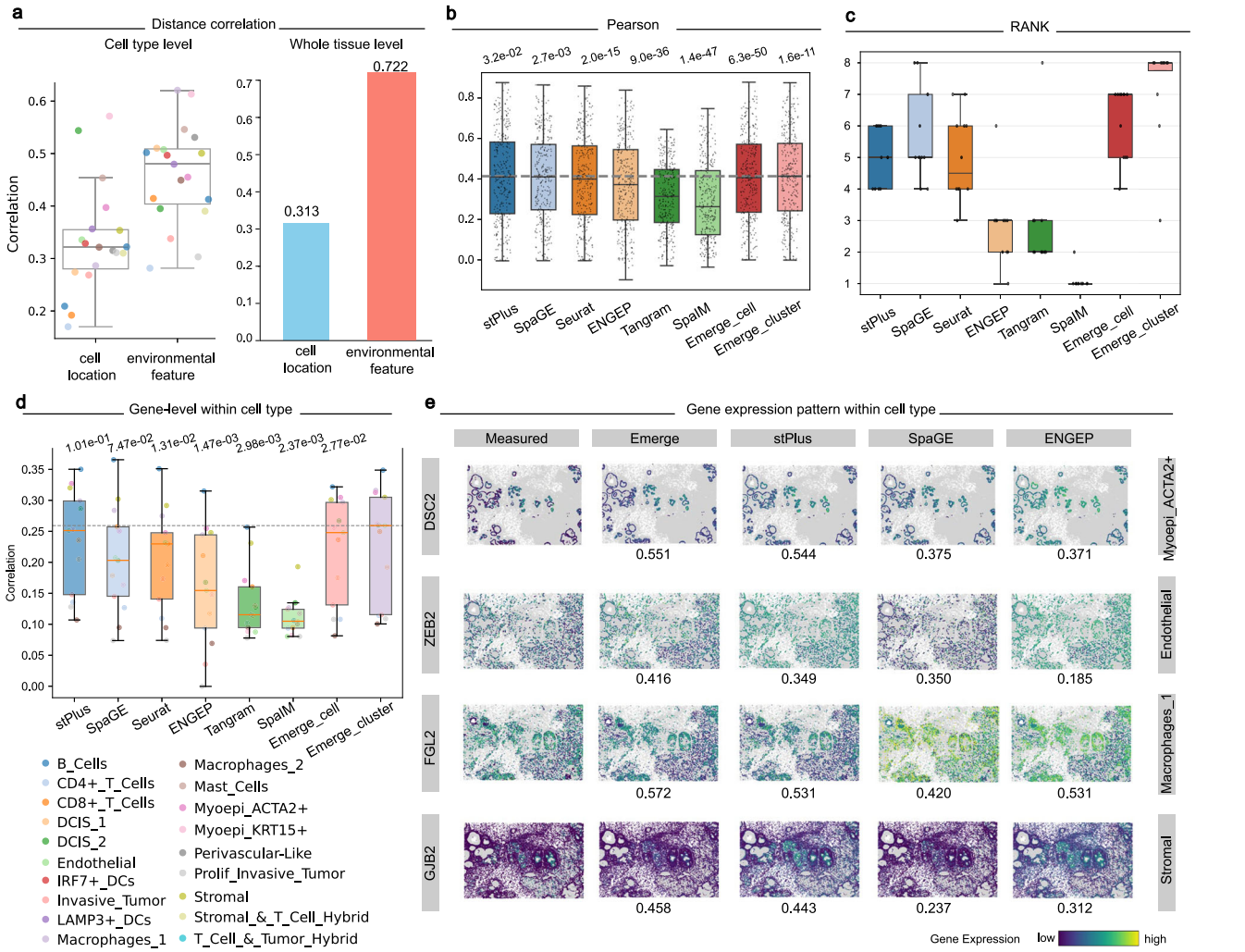


Fig. 4 | Microenvironment-aware reconstruction captures structured transcriptional variation in breast cancer. a Distance correlations between gene expression and either cell location or environmental features. Correlations are calculated across individual cell types ($n = 19$ cell types; left, each point represents one cell type) and across all cells in the tissue (right). Numbers above the bars indicate the corresponding distance correlation values. **b** Gene-level reconstruction performance measured by Pearson correlation between predicted and measured expression across genes. Each box summarizes the distribution of gene-wise Pearson correlations ($n = 306$ genes) for each method. Box plots show median values, interquartile ranges, and whiskers extending to 1.5 × IQR. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests, with exact p values shown above the box plots. **c** AR (aggregated accuracy rank) scores integrating gene-level and cell-level reconstruction performance across six evaluation metrics. Higher AR scores indicate better overall performance. **d** Within-cell-type reconstruction performance evaluated using structured genes. Pearson correlations between predicted and measured expression are shown for each method. Each dot represents one cell type ($n = 19$ cell types), with values corresponding to the mean gene-wise Pearson correlation across structured genes. Colors denote different cell types. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests. **e** Spatial expression patterns of representative genes within selected cell types. Measured expression and reconstructed expression from different methods are shown side by side. Numbers indicate Pearson correlations between predicted and measured expression. Source data are provided as a Source Data file.

more faithfully recovered the spatial confinement and cell-type specificity observed in the measured data, whereas competing approaches frequently produced diffuse or fragmented patterns.

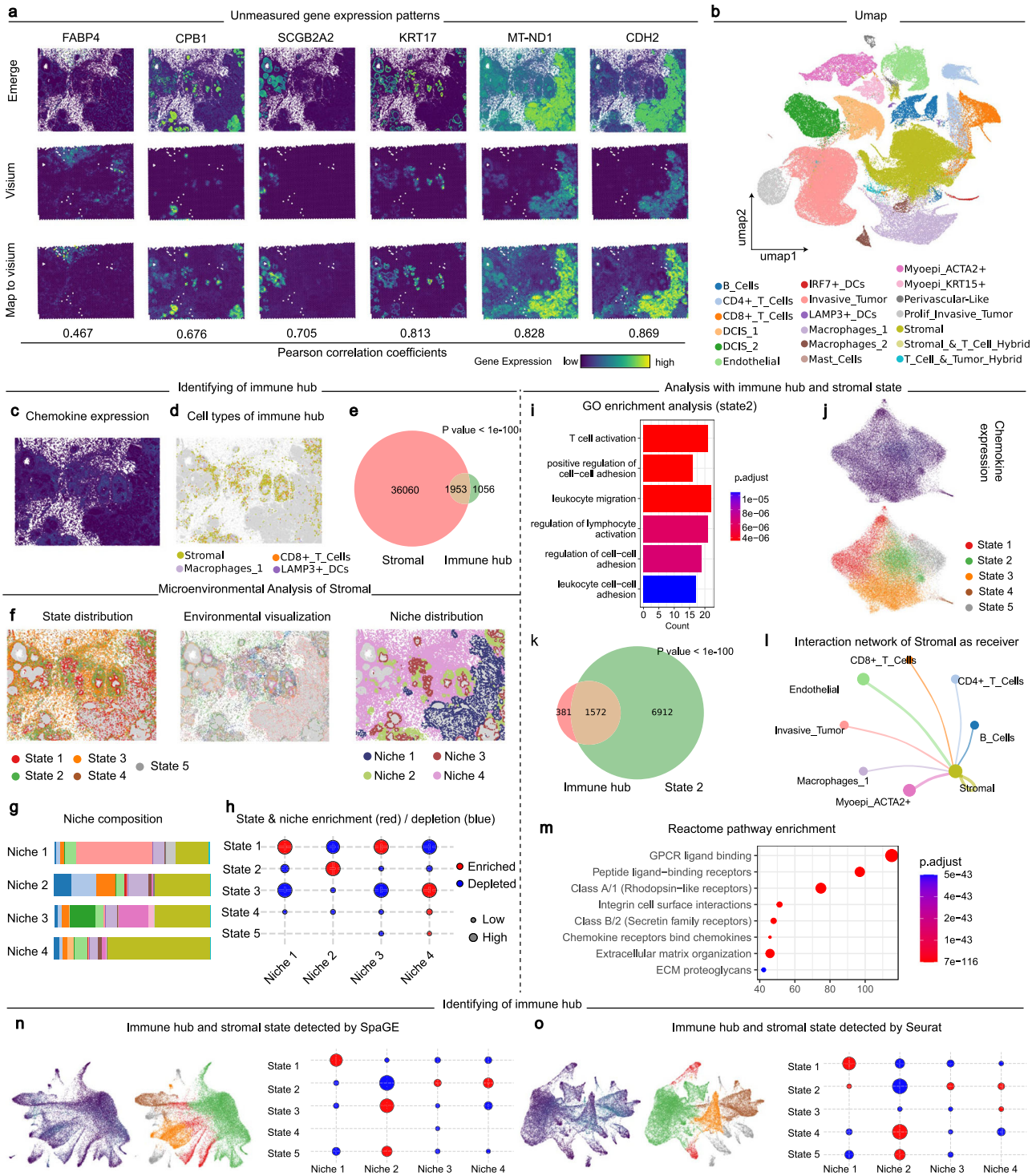
Together, these results indicate that local microenvironmental organization constitutes a major source of transcriptional variation in breast tumors. By integrating transcriptomic similarity with niche-associated relational information, Emerge improves recovery of structured cellular variation associated with local cellular interactions and context-dependent tumor states.

Emerge uncovers chemokine-defined immune stromal hubs absent from targeted panels

We next asked whether transcriptome-scale reconstruction could reveal biologically important programs that are not directly measurable by targeted spatial transcriptomics panels. To first validate the reconstruction of genes absent from the Xenium assay, we compared reconstructed expression with matched Visium profiles from an adjacent tissue section. Predicted single-cell expression was aggregated into pseudo-Visium spots for evaluation. Across genes, Emerge achieved the highest median gene-wise correlation among competing methods and faithfully recovered representative unmeasured genes (Fig. 5a and Supplementary Fig. 24a). Incorporating reconstructed genes further expanded transcriptional resolution beyond the measured panel while preserving major cell identities and global tissue organization (Fig. 5b and Supplementary Fig. 24b, c).

Having established reconstruction accuracy, we next investigated whether previously unmeasured genes could reveal novel biological programs. Notably, several key chemokines involved in antitumor immunity, including *CXCL9*, *CXCL10* and *CXCL11*[53–55], are absent from the Xenium panel yet were robustly recovered by Emerge. Their

https://doi.org/10.1038/s41467-026-77349-8



reconstructed expression defined discrete regions of elevated chemokine activity, hereafter referred to as immune hubs (Fig. 5c). Strikingly, these immune hubs were predominantly localized within stromal-rich rather than tumor-dominated regions (Fig. 5d, e and Supplementary Fig. 24d), suggesting the existence of specialized stromal programs associated with local immune organization[56].

To investigate this possibility, we characterized stromal heterogeneity using reconstructed transcriptomes together with local environmental features. Five stromal states were identified across four microenvironmental niches (Fig. 5f and Supplementary Fig. 24e–i), among which one population (State 2) emerged as a distinct immune-associated stromal program. State 2 was preferentially enriched within an immune-rich niche (Niche 2) containing abundant B cells, $CD4^+$ T cells, $CD8^+$ T cells and macrophages (Fig. 5g, h) and exhibited strong enrichment for T-cell activation, lymphocyte activation, leukocyte migration and immune-cell adhesion pathways, distinguishing it from the remaining stromal states, which were associated with diverse extracellular matrix, developmental, metabolic and immune-related programs[57] (Fig. 5i and Supplementary Fig. 25a, b and Supplementary Data 2).

Remarkably, reconstructed immune hubs showed strong spatial overlap with State 2 stromal cells (Fig. 5j, k). Statistical analyses

Fig. 5 | Immune-associated stromal programs revealed by transcriptome-scale reconstruction in breast cancer. a Spatial expression patterns of representative unmeasured genes reconstructed by Emerge. Top: Xenium cell-level predictions. Middle: measured Visium expression from adjacent sections. Bottom: pseudo-Visium spot profiles aggregated from reconstructed cell-level predictions. Numbers indicate Pearson correlations between pseudo-Visium and measured Visium expression. **b** UMAP embedding of spatial cells based on combined measured and reconstructed gene expression. **c**–**e** Identification of chemokine-defined immune hubs. **c** Spatial distribution of reconstructed chemokine activity. **d** Cell-type composition of immune hub regions. **e** Overlap between immune hubs and stromal cells, with *p* value assessed by hypergeometric enrichment test. **f**–**h** Microenvironmental analysis of stromal cells. **f** Spatial distribution of stromal states (left), environmental visualization of stromal cells (middle), and spatial distribution of microenvironmental niches (right). **g** Cell-type composition of each niche. **h** Enrichment and depletion of stromal states across niches; red dots indicate enrichment, blue dots indicate depletion, and dot size reflects statistical significance. **i**–**m** Characterization of immune-associated stromal State 2. **i** Gene Ontology enrichment analysis of marker genes of stromal State 2. Enrichment significance was assessed by over-representation analysis with Benjamini-Hochberg adjusted *p* values. **j** UMAP projections showing reconstructed immune chemokine expression (top) and stromal-state distribution (bottom). **k** Overlap between immune hubs and stromal State 2, with *p* value assessed by hypergeometric enrichment test. **l** Ligand-receptor interaction network inferred by CellChat with stromal cells treated as receivers. **m** Reactome pathway enrichment analysis of inferred ligands and receptors. Enrichment significance was assessed by over-representation analysis with Benjamini-Hochberg adjusted *p* values. **n** Results from SpaGE: reconstructed immune chemokine expression (left), stromal-state distribution (middle), and enrichment/depletion of stromal states across niches (right). **o** Results from Seurat: reconstructed immune chemokine expression (left), stromal-state distribution (middle), and enrichment/depletion of stromal states across niches (right). Source data are provided as a Source Data file.

confirmed significant enrichment of immune hubs within this state, whereas the remaining stromal populations exhibited little or no association[56] (Supplementary Fig. 25c). Cell-cell communication analysis further demonstrated that State 2 engaged with T cells, B cells, macrophages, endothelial cells and neighboring tumor populations within the local immune ecosystem (Fig. 5l). Correspondingly, receptor-mediated communication was enriched for GPCR-mediated signaling, peptide ligand binding, chemokine receptor interactions and integrin cell-surface pathways (Fig. 5m), suggesting that this stromal population may contribute to local immune organization through coordinated chemokine and receptor-mediated communication programs.

Finally, we examined whether comparable immune-associated stromal programs could be recovered by alternative reconstruction methods. We focused on SpaGE[14] and Seurat[13] because they were among the strongest-performing baseline approaches in our reconstruction benchmarks (Fig. 4b, d). Although both methods identified stromal populations and regions of elevated immune activity, the correspondence between stromal states and reconstructed immune hubs was substantially weaker, with immune hubs frequently fragmented across multiple stromal populations (Fig. 5n, o and Supplementary Fig. 26). In contrast, Emerge consistently recovered a coherent immune-associated stromal state exhibiting strong and specific enrichment of reconstructed chemokine activity.

Together, these findings indicate that transcriptome-scale reconstruction extends the biological scope of imaging-based spatial transcriptomics beyond the measured gene panel. By recovering previously unmeasured chemokine programs and linking them to a discrete immune-associated stromal state, Emerge uncovers a layer of microenvironmental organization that would otherwise remain inaccessible from panel-limited spatial transcriptomics.

Microenvironment-aware reconstruction generalizes across diverse metastatic breast cancer ecosystems

To determine whether microenvironment-aware reconstruction generalizes across distinct metastatic ecosystems, we analyzed twelve metastatic breast cancer (MBC) sections profiled by MERFISH together with matched sc/snRNA-seq reference from adjacent sections[58]. Despite substantial variation in cellular composition, tissue architecture and epithelial-mesenchymal transition (EMT) status across lesions, local environmental features exhibited stronger associations with gene expression than spatial coordinates at the whole-tissue level and across the majority of major cell types (Fig. 6a and Supplementary Figs. 27 and 28). These observations indicate that local cellular organization represents a conserved determinant of transcriptional variation across diverse metastatic ecosystems.

We next evaluated whether this conserved environmental information could be exploited for transcriptome reconstruction. Across five-fold cross-validation analyses, Emerge_cell and Emerge_cluster consistently ranked among the best-performing reconstruction methods at both the gene and cell levels across metastatic lesions despite pronounced inter-sample heterogeneity (Fig. 6b, c and Supplementary Figs. 29–41). Moreover, incorporating microenvironmental information consistently improved reconstruction relative to transcriptome-only alignment across lesions (Supplementary Figs. 42 and 43), demonstrating that environmental context provides complementary information beyond intrinsic transcriptomic similarity across metastatic lesions.

Consistent with observations in the mouse cortex and primary breast cancer datasets, the largest and most reproducible gains were concentrated in structured genes exhibiting both substantial within-cell-type variability and spatial organization (Fig. 6d and Supplementary Fig. 44). Similar trends were observed for HVGs, SVGs and marker genes, whereas improvements were markedly attenuated for non-HVG and non-SVG control genes (Supplementary Figs. 45–49), indicating that local environmental information preferentially enhances recovery of structured transcriptional programs associated with cellular states rather than uniformly improving all genes.

Representative examples further illustrated these advantages. At the tissue level, Emerge accurately recovered spatial expression programs associated with fibroblast- and immune-enriched regions (Supplementary Fig. 50). More importantly, within fibroblast populations, Emerge faithfully reconstructed localized expression patterns of *COL1A2*, *MYL9* and *CD163*, preserving fine-grained spatial heterogeneity that was frequently oversmoothed or fragmented by competing approaches[59–61] (Fig. 6e and Supplementary Fig. 51).

Collectively, these findings indicate that microenvironment-aware reconstruction generalizes robustly across metastatic breast cancer ecosystems. Despite extensive variation in tissue architecture and disease state, local environmental organization remains a conserved source of structured transcriptional variation that can be systematically leveraged to improve transcriptome reconstruction across diverse metastatic lesions.

Emerge resolves EMT-associated fibroblast remodeling across metastatic breast cancer ecosystems

We next asked whether transcriptome-scale reconstruction could uncover conserved fibroblast remodeling programs associated with epithelial-mesenchymal transition (EMT) across metastatic breast cancer ecosystems. Integrating reconstructed fibroblasts from twelve metastatic lesions resolved five reproducible fibroblast states, whereas measured genes alone separated only two broad populations (Fig. 7a, b and Supplementary Fig. 52a–c), substantially increasing the resolution of fibroblast heterogeneity.

Despite pronounced inter-lesion heterogeneity, fibroblast-state composition exhibited striking EMT-associated remodeling (Fig. 7c).

https://doi.org/10.1038/s41467-026-77349-8

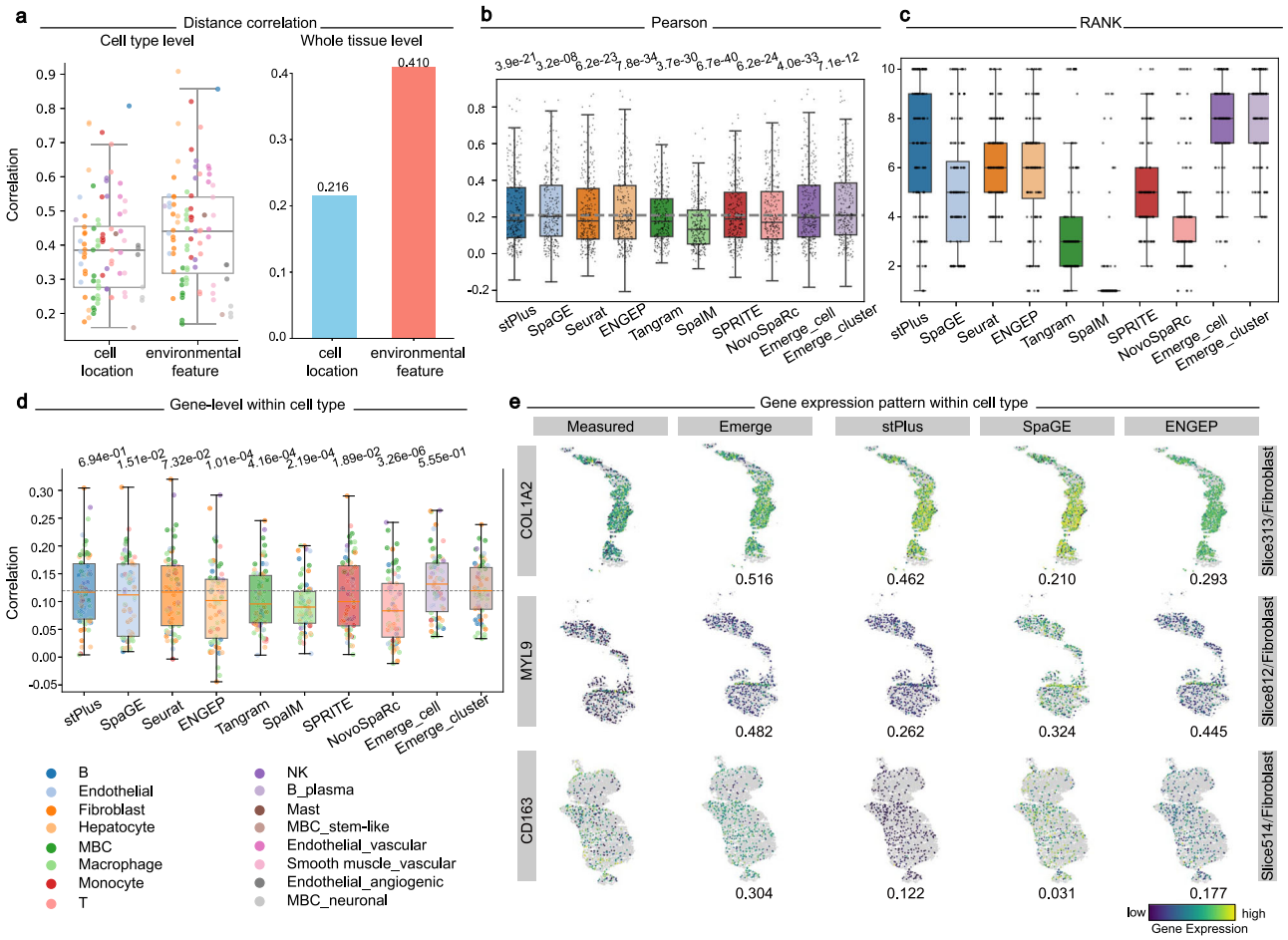


Fig. 6 | Microenvironment-aware reconstruction generalizes across metastatic breast cancer ecosystems. a Distance correlations between gene expression and either cell location or environmental features across 12 metastatic breast cancer (MBC) tissue sections. Correlations are summarized across individual cell types ($n = 88$ slice-specific cell-type instances; left, each point represents one slice-specific cell-type instance) and across all cells (right). Numbers above the bars indicate the corresponding distance correlation values. **b** Gene-level reconstruction performance in a representative metastatic section (slice 514), measured by Pearson correlation between predicted and measured expression across genes. Each box summarizes the distribution of gene-wise Pearson correlations ($n = 286$ genes) for each method. Box plots show median values, interquartile ranges and whiskers extending to 1.5 × IQR. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests, with exact p values shown above the box plots. **c** AR (aggregated accuracy rank) scores integrating gene-level and cell-level reconstruction performance across 12 MBC tissue sections using six evaluation metrics. Higher AR scores indicate better overall performance. **d** Within-cell-type reconstruction performance evaluated using structured genes across MBC tissue sections. Pearson correlations between predicted and measured expression are shown for each method. Each point represents one slice-specific cell-type instance ($n = 88$ instances), with values corresponding to the mean gene-wise Pearson correlation across structured genes. Colors denote cell types. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests. **e** Spatial expression patterns of representative genes within fibroblast populations. Measured expression and reconstructed expression from different methods are shown side by side. Numbers indicate Pearson correlations. Source data are provided as a Source Data file.

State 2 predominated in EMT-low lesions, whereas State 3 became the dominant fibroblast program in EMT-high lesions. EMT-patched lesions displayed more heterogeneous compositions without a single prevailing state. EMT scores calculated from reconstructed transcriptomes closely matched those derived from measured genes (Supplementary Fig. 53), supporting the biological validity of the inferred programs. Functional analysis further distinguished the two dominant EMT-associated states: State 2 was enriched for biosynthetic and secretory processes, whereas State 3 was characterized by extracellular matrix organization and collagen remodeling (Fig. 7d and Supplementary Fig. 52d and Supplementary Data 3). Representative marker genes further illustrated their distinct spatial distributions across metastatic lesions (Fig. 7e and Supplementary Figs. 54–58). Together, these findings reveal reproducible EMT-associated fibroblast remodeling, marked by reduced dominance of a biosynthetic program and increased dominance of a matrix-remodeling program.

We next asked whether these EMT-associated fibroblast programs occupied distinct microenvironmental niches. Clustering environmental feature vectors identified five ecological niches with characteristic cellular compositions (Fig. 7f, g and Supplementary Fig. 59), and fibroblast states exhibited pronounced niche preferences (Fig. 7h). Specifically, the biosynthetic program (State 2) was preferentially enriched in endothelial- and macrophage-rich niches, whereas the matrix-remodeling program (State 3) occupied vascular-associated niches containing abundant immune populations[62]. Consistent with these niche preferences, CellChat[63] analysis revealed extensive rewiring of intercellular communication programs (Fig. 7i–k and Supplementary Fig. 52e–g). Relative to State 2, State 3 displayed increased activity of integrin-family receptors together with stronger enrichment of chemokine-, interleukin- and GPCR-mediated signaling pathways, consistent with enhanced extracellular matrix interactions and immune-associated communication.

https://doi.org/10.1038/s41467-026-77349-8

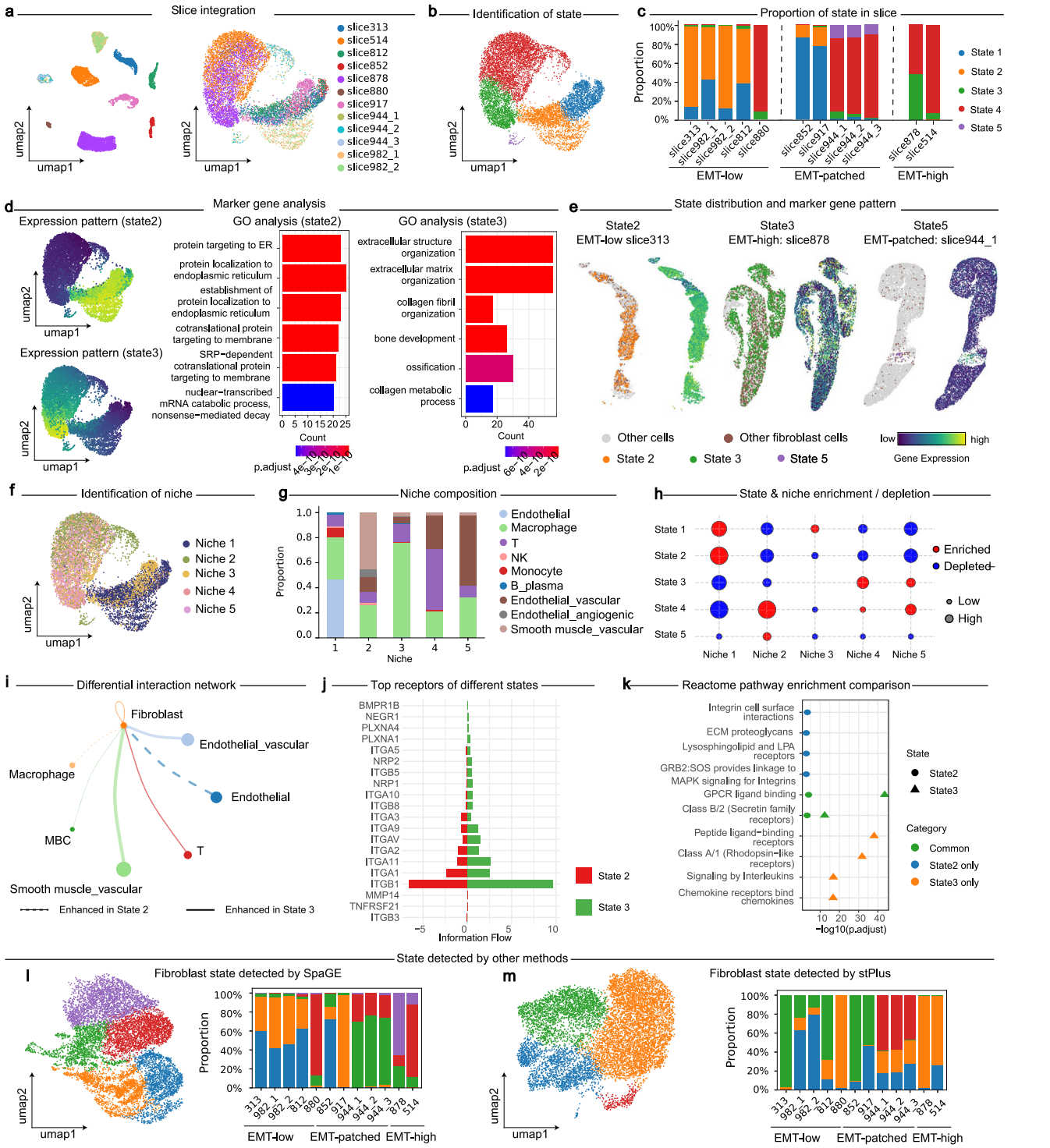


Fig. 7 | Transcriptome-scale reconstruction reveals EMT-associated fibroblast remodeling across metastatic breast cancer ecosystems. a UMAP of fibroblasts before (left) and after (right) integration across 12 sections using BBKNN (batch balanced k nearest neighbors), based on combined measured and reconstructed gene expression. **b** Fibroblast states identified by Louvain clustering of the integrated dataset. **c** Distribution of fibroblast states across 12 sections, grouped by EMT (Epithelial-Mesenchymal Transition) phenotype (EMT-low, EMT-patched and EMT-high). **d** Marker gene expression (left; State 2, top; State 3, bottom) and Gene Ontology enrichment analysis of marker genes. Enrichment significance was assessed by over-representation analysis with Benjamini-Hochberg adjusted *p* values. **e** Spatial distribution of fibroblast states (left) and representative marker expression (right) in sections from each EMT group. **f** Hierarchical clustering of fibroblast environmental feature vectors defining five niche categories. **g** Average cellular composition of each niche. **h** Enrichment (red dots) and depletion (blue dots) of cell states across niches assessed by hypergeometric tests with Benjamini-Hochberg FDR correction. Dot size represents $-\log_{10}$(adjusted *p* value). **i** Differential intercellular interaction networks between fibroblast states and neighboring cell populations inferred by CellChat. **j** Top receptor genes contributing to differential signaling across fibroblast states. **k** Reactome pathway enrichment analysis of inferred ligands and receptors. Enrichment significance was assessed by over-representation analysis with Benjamini-Hochberg adjusted *p* values. **l**, **m** Fibroblast states identified by alternative reconstruction methods. **l** SpaGE: UMAP of fibroblast states (left) and their distribution across EMT phenotypes (right). **m** stPlus: UMAP of fibroblast states (left) and their distribution across EMT phenotypes (right). Source data are provided as a Source Data file.

Finally, we examined whether these coordinated biological patterns could be recovered by alternative transcriptome-completion methods. Although SpaGE[14] and stPlus[16] partially recapitulated fibroblast heterogeneity, both failed to reproducibly recover the coordinated associations among fibroblast states, EMT context and ecological niches observed with Emerge (Fig. 7l, m and Supplementary Fig. 60). In contrast, Emerge consistently identified coherent EMT-associated fibroblast remodeling linked to microenvironmental organization across metastatic lesions.

Collectively, these findings indicate that transcriptome-scale reconstruction enables systematic characterization of EMT-associated fibroblast remodeling across metastatic ecosystems. By recovering microenvironment-associated transcriptional variation beyond targeted panels, Emerge reveals coordinated state-niche relationships that remain largely inaccessible using measured genes alone.

Discussion

A central challenge in spatial biology is that cells sharing the same identity can occupy distinct transcriptional states depending on the local environments in which they reside. This challenge is particularly pronounced in imaging-based spatial transcriptomics, where targeted gene panels are often sufficient to identify major cell types but insufficient to capture the full spectrum of context-dependent transcriptional variation. Existing transcriptome-completion approaches largely rely on shared-gene alignment between spatial and reference datasets, implicitly assuming that transcriptional similarity alone is sufficient to recover unmeasured expression. Our results suggest that this view is incomplete. Across multiple tissues, disease contexts and imaging-based spatial transcriptomics platforms, local microenvironmental composition provided information complementary to shared-gene similarity and was generally more strongly associated with transcriptional variation than spatial proximity. Building on this observation, Emerge extends transcriptome reconstruction as a context-aware modeling problem in which cellular states are jointly constrained by intrinsic identity and local microenvironment context.

The central insight underlying Emerge is that cell identity and cellular context capture complementary aspects of biological variation. Cell identity defines the broad transcriptional manifold within which a cell resides, whereas local microenvironments are associated with state-specific programs. Across all datasets examined, cells sharing similar identities but residing in distinct microenvironment frequently exhibited different transcriptional programs. These observations suggest that cell identity alone is insufficient to fully explain cellular states and that local tissue organization contains biologically meaningful information complementary to transcriptomic similarity. By integrating a first-order expression objective with a second-order environmental objective, Emerge preserves global identity structure while recovering context-dependent variation associated with local niches. Together, these findings highlight the complementary roles of intrinsic cell identity and extrinsic tissue context linked to context-dependent cellular states.

Beyond improving transcriptome reconstruction, microenvironment-aware modeling enabled recovery of biologically meaningful state variation that remained largely inaccessible from measured genes alone. Across multiple tissues and disease contexts, reconstructed transcriptomes consistently revealed coordinated relationships among cellular states, ecological niches and intercellular communication programs, suggesting that local microenvironments provide a recurring organizing principle for context-dependent transcriptional variation. More broadly, these findings suggest that transcriptome reconstruction should be viewed not only as a gene-completion problem but also as a means of recovering higher-order biological organization within complex tissue ecosystems.

The benefits of microenvironment-aware reconstruction remained robust across alternative annotation strategies, microenvironmental feature definitions and reference datasets. Importantly, environmental modeling remained beneficial even under fine-grained subtype annotations and substantially reduced the dependence of reconstruction accuracy on annotation granularity, indicating that local niche information captures biological variation that is complementary to, rather than redundant with, increasingly refined cell identities. Moreover, the framework remained effective across references derived from different cohorts, platforms and experimental settings, suggesting that the contribution of microenvironmental modeling reflects a reproducible biological signal rather than dependence on a particular reference or implementation. Together, these observations indicate that microenvironment-aware reconstruction is robust to realistic levels of reference heterogeneity and does not require strict sample-level correspondence. Additional analyses related to computational scalability are provided in Supplementary Note 2.7.

Despite these encouraging results, several limitations remain. As with all reference-based approaches, reconstruction quality depends on the coverage and quality of the reference atlas, and cellular states absent from the reference remain difficult to recover. In addition, Emerge represents tissue context through neighboring cell-type composition, which is interpretable and robust but captures only one aspect of the local microenvironment. Future extensions could incorporate additional contextual information, including cellular morphology, tissue architecture and molecular interaction networks. Performance may also be influenced by platform-specific factors, including segmentation quality, transcript-assignment uncertainty, background noise and spatial resolution, particularly when applying Emerge to emerging platforms that require additional cell segmentation or transcript-assignment procedures. Finally, Emerge identifies associations between cellular states and local niches rather than causal relationships and should therefore primarily be viewed as a hypothesis-generating framework requiring downstream experimental validation.

In summary, Emerge extends transcriptome reconstruction from recovering missing genes to recovering missing biological context. Our results suggest that cellular states cannot be fully understood through cell identity alone but must be interpreted within the ecological environments in which they arise. By explicitly modeling local microenvironments, Emerge enables recovery of context-dependent transcriptional programs that are largely inaccessible from targeted measurements alone. As spatial atlases continue to expand in scale and resolution, integrating intrinsic cell identity with extrinsic tissue context will be essential for understanding how cellular states emerge, diversify and function within complex biological systems.

Methods

Emerge is a computational framework that extends imaging-based spatial transcriptomics from targeted gene panels to transcriptome-scale resolution. The framework integrates spatial gene expression $\mathbf{X}_q$ and spatial coordinates $\mathbf{Z}_q$, obtained from upstream segmentation and transcript-assignment pipelines, with annotated sc/snRNA-seq reference profiles $\mathbf{X}_r$ and cell-type labels $\mathbf{Y}_r$. Emerge comprises a core inference pipeline for confidence-aware cell-type annotation and microenvironment-aware transcriptome reconstruction, together with a downstream analysis module for characterizing state-niche relationships and microenvironment-associated cellular programs (Fig. 1). A summary of notations is provided in Supplementary Note 3.1.

Core Emerge framework

Emerge is built on the premise that cell identity and local microenvironment provide complementary constraints on cellular state. Cell identity constrains the broad transcriptional manifold within which a

cell resides, whereas microenvironmental context refines variation within that manifold. To integrate these complementary sources of information, Emerge comprises two interconnected modules: a Cell Type Annotator and a Gene Expression Predictor (detailed algorithmic descriptions are provided in Supplementary Notes 3.2 and 3.3, respectively). Together, they enable confidence-aware cell-type annotation and microenvironment-aware transcriptome reconstruction.

Cell type annotator. Accurate cell-type annotation in imaging-based ST is challenging due to limited gene panels, measurement noise, and potential mismatch between spatial data and reference atlases. In particular, not all spatial cells are equally well represented in the reference: while some exhibit canonical profiles, others may correspond to rare, transitional, or context-specific states. Treating all cells as equally well aligned can therefore propagate annotation errors into downstream reconstruction.

To address this, Emerge employs an ensemble k-nearest neighbors (kNN) annotator that integrates multiple similarity metrics and neighborhood sizes in the shared measured-gene space. For each query cell q_i, each metric-neighborhood configuration (m, k) defines a neighbor set $NN^{(m,\,k)}(q_i)$ in the reference and induces a provisional probability distribution over cell types:

$$\hat{\mathbf{Y}}_{q_i}^{(m,k)} = \sum_{r_j \in NN^{(m,k)}(q_i)} w^{(m,k)}(q_i, r_j)\mathbf{Y}_{r_j}, \tag{1}$$

where $w^{(m,k)}(q_i, r_j)$ are normalized similarity weights and $\mathbf{Y}_{r_j}$ is the one-hot type vector

$$\mathbf{Y}_{q_i} = \frac{1}{|\mathcal{M}||\mathcal{K}|} \sum_{m\in\mathcal{M}} \sum_{k\in\mathcal{K}} \hat{\mathbf{Y}}_{q_i}^{(m,k)}. \tag{2}$$

where $\mathcal{M}$ denotes the set of all metric-neighborhood configurations.

We quantify annotation reliability using the maximum assignment probability,

$$f_{q_i} = \max(\mathbf{Y}_{q_i}) \tag{3}$$

The confidence score is therefore defined as the maximum consensus assignment probability produced by the ensemble annotation framework. A threshold of 0.5 is used to define high-confidence cells, indicating that a cell is assigned to a given cell type with greater probability than all alternative identities combined. This threshold provides a simple and interpretable criterion for distinguishing reliable assignments from uncertain annotations. Similar consensus-based strategies have previously been used to improve annotation robustness and assess annotation reliability in single-cell annotation studies[64,65].

Based on this confidence measure, Emerge separates high-confidence and low-confidence cells. Only high-confidence cells participate in the subsequent type-constrained FGW (fused Gromov-Wasserstein) alignment, thereby reducing the propagation of uncertain annotations into downstream reconstruction. High-confidence cells are assigned the type $\arg\max(Y_{q_i})$ and used as anchors for downstream reconstruction, whereas low-confidence cells are excluded from direct alignment and handled separately through neighborhood-guided reconstruction. This design reduces the influence of uncertain or potentially mismatched annotations while retaining information from ambiguous or transitional cellular states in the final transcriptome reconstruction.

This confidence-aware annotation step establishes a probabilistic identity backbone for spatial cells. When spatial cluster annotations are available, Emerge optionally operates in a cluster-level mode (Emerge_cluster), in which cell-type probabilities are aggregated within each cluster, and a consensus label is assigned to improve robustness in noisy or low-coverage datasets. Otherwise, Emerge operates in single-cell mode (Emerge_cell). In this study, Emerge_cluster was used whenever platform-provided spatial clusters were available, whereas Emerge_cell was retained as a single-cell-resolution variant for benchmarking analyses.

Gene expression predictor. Reconstructing unmeasured genes requires modeling both intrinsic transcriptional similarity and local microenvironmental context. Existing approaches primarily rely on shared-gene alignment, whereas Emerge additionally incorporates relational information encoded by local tissue organization. To achieve this, Emerge establishes a correspondence between transcriptomic organization in the reference and microenvironmental structure in spatial data. The reconstruction framework consists of three components: (1) adaptive microenvironment characterization, (2) type-constrained FGW-OT for high-confidence cells, and (3) uncertainty-aware regression for low-confidence cells.

Environmental feature construction. For each spatial cell q_i, the microenvironment is defined as a continuous descriptor based on the local composition of neighboring cell types:

$$\mathbf{E}(q_i) = \sum_{q_{i'}\in\mathcal{N}(q_i)} w_{ii'}\mathbf{Y}_{q_{i'}}. \tag{4}$$

where $\mathcal{N}(q_i)$ denotes the set of spatial neighbors defined by k-nearest neighbors in coordinate space, and $w_{ii'}$ are normalized weights derived from a Gaussian kernel:

$$w_{ii'} \propto \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_{i'}\|_2^2}{2\sigma_i^2}\right), \tag{5}$$

with bandwidth σ_i adapted to local cell density. Self-contributions are excluded to prevent trivial self-reinforcement.

This adaptive formulation yields a continuous representation of local tissue context that remains comparable across regions with heterogeneous spatial densities. Neighboring cell-type composition is used as the default environmental representation because of its biological interpretability and direct connection to local cellular organization. However, the FGW framework itself is agnostic to the specific choice of environmental descriptor and can accommodate alternative representations of local context, including neighborhood embeddings and graph-derived features as used in previous studies[45,46]. To assess the dependence of Emerge on environmental feature definition, we systematically evaluated multiple neighborhood constructions and alternative environmental representations (Supplementary Note 2.4).

Type-constrained FGW-OT for high-confidence cells. For each cell type c, Emerge aligns high-confidence spatial cells to reference cells of the same annotated type, preserving global identity structure while capturing context-dependent variation within each type. Let $\mathbf{X}_q^{(c)}$ and $\mathbf{X}_r^{(c)}$ denote the shared-gene expression matrices of spatial and reference cells belonging to cell type c, respectively.

The transport coupling $\mathbf{T}^{(c)}$ is estimated by solving the entropically regularized fused Gromov-Wasserstein problem:

$$\mathbf{T}^{(c)} = \arg\min_{\mathbf{T}\in\Pi} (1-\alpha)D_1^{(c)}(\mathbf{T}) + \alpha D_2^{(c)}(\mathbf{T}) - \epsilon H(\mathbf{T}), \tag{6}$$

where Π denotes the set of admissible couplings with prescribed marginals, and $H(\mathbf{T}) = -\sum_{i,j} T_{ij}\log T_{ij}$ is the entropy regularizer.

The first-order term aligns spatial and reference cells according to shared-gene expression similarity:

$$D_1^{(c)}(\mathbf{T}) = \sum_{i,j} C_{ij}^{(c)} T_{ij}, \tag{7}$$

where $C_{ij}^{(c)}$ denotes the normalized distance between spatial cell q_i and reference cell r_j, computed using multiple similarity metrics and averaged to improve robustness. The second-order term aligns relational structures between the spatial microenvironment and the reference transcriptomic manifold:

$$D_2^{(c)}(\mathbf{T}) = \sum_{i,i'} \sum_{j,j'} \frac{\left(D_{ii'}^{\mathrm{env}(c)} - D_{jj'}^{\exp(c)}\right)^2}{2} T_{ij} T_{i'j'}, \tag{8}$$

where $D_{ii'}^{\mathrm{env}(c)}$ denotes pairwise distances between environmental descriptors of spatial cells q_i and $q_{i'}$, and $D_{jj'}^{\exp(c)}$ denotes pairwise transcriptomic distances between reference cells r_j and $r_{j'}$. Both distance matrices are normalized prior to alignment.

The first-order term provides the primary correspondence between spatial and reference cells based on measured gene expression, whereas the second-order term introduces a relational constraint that encourages consistency between similarities in spatial microenvironments and similarities in the reference transcriptomic manifold. Unlike direct matching objectives, this formulation does not require explicit correspondence between individual environmental states and reference cell states. Instead, it preserves relative organization among cells, allowing microenvironmental information to refine alignment while remaining anchored by transcriptomic similarity. Together, the two objectives jointly constrain alignment through intrinsic expression similarity and local tissue context, enabling reconstruction of transcriptomic variation associated with microenvironmental organization within cell types.

The parameter α controls the relative contribution of the first-order transcriptomic objective and the second-order environmental objective. Unless otherwise specified, we use $\alpha = 0.5$, which provides balanced contributions from the two objectives. Sensitivity analyses across multiple datasets demonstrated stable performance across a broad range of values, with intermediate values consistently achieving near-optimal reconstruction (Supplementary Note 2.2, Supplementary Figs. 3, 15 and 42).

The optimized coupling is then used to reconstruct transcriptome-scale expression for high-confidence cells:

$$\widehat{\mathbf{X}}_{q,\mathrm{high}}^{(c)} = \mathbf{T}^{(c)} \mathbf{X}_r^{(c)} \tag{9}$$

Regression-based imputation for low-confidence cells. Low-confidence cells are not directly aligned because their identities are considered uncertain. Instead, their transcriptomes are inferred using an ensemble kNN regression anchored on nearby high-confidence cells. Distances are computed in the shared-gene expression space and predictions are aggregated across multiple neighborhood sizes. This strategy prevents uncertain annotations from directly influencing the FGW alignment while still allowing ambiguous or transitional cells to contribute to the final reconstructed transcriptome.

Microenvironment-aware spatial analyzer

Building upon the reconstructed transcriptomes and microenvironmental descriptors, Emerge provides a unified framework for characterizing how cellular states relate to local tissue niches and higher-order spatial organization. The framework consists of three components: (1) identification of microenvironment-associated cellular states, (2) characterization of state-niche architecture within tissues, and (3) integration of state-niche relationships across samples.

Identification of microenvironment-associated cellular states. Within each broad cell type, cellular states are identified by clustering reconstructed full-transcriptome profiles. For each state, the distribution of environmental descriptors is quantified and compared across states to evaluate enrichment for specific microenvironmental contexts. Differential expression and pathway enrichment analyses are subsequently performed to characterize state-specific molecular programs. This analysis enables systematic identification of microenvironment-associated cellular states and their corresponding transcriptional programs within each cell type.

Characterization of state-niche architecture. To characterize mesoscale tissue organization, Emerge jointly analyzes reconstructed cellular states and environmental descriptors across spatial locations. Cells sharing similar transcriptional programs and microenvironmental contexts are aggregated into spatially coherent regions, enabling characterization of region-specific cellular composition, pathway activity, and intercellular communication patterns. This analysis links cellular states to their surrounding niches and facilitates the identification of spatially organized biological programs within tissues.

Integration of state-niche relationships across samples. For multi-section or multi-sample datasets, Emerge aligns cellular states and environmental descriptors across samples to enable cross-sample comparison. This alignment enables identification of conserved and context-specific state-niche relationships, facilitating comparison of how microenvironment-associated cellular programs are maintained or rewired across biological conditions.

Implementation details for clustering, enrichment analysis, communication inference, and cross-sample alignment are provided in Supplementary Note 3.4.

Dataset

All datasets analyzed in this study are publicly available. We evaluated Emerge across three representative imaging-based ST datasets spanning distinct platforms, panel sizes, and biological contexts (Supplementary Data 4). These include: (1) a MERFISH dataset of the mouse primary motor cortex (MOp)[41], providing high cellular diversity and well-characterized spatial lamination; (2) a Xenium dataset of human breast cancer tissue[4], characterized by pronounced microenvironmental heterogeneity and immune infiltration; and (3) 12 MERFISH sections from metastatic breast cancer (MBC) patients[58], enabling cross-section and cross-patient analyses of stromal and immune programs.

Reference datasets included both matched and independent references. Matched references refer to biologically compatible datasets derived from the same tissue context and sharing broadly consistent cellular compositions, but do not require the spatial and reference datasets to originate from the same biological sample. Independent references refer to scRNA-seq or snRNA-seq datasets generated from separate cohorts, experiments, or sequencing technologies, in which major cell-type correspondence is preserved but fine-grained cellular states may differ. Detailed descriptions of datasets, reference configurations, and preprocessing procedures are provided in Supplementary Note 4.

In addition to the primary references used for transcriptome reconstruction, we further evaluated robustness using multiple independent reference datasets spanning distinct biological cohorts, sequencing technologies and experimental settings. These analyses were designed to assess the sensitivity of Emerge to realistic levels of reference heterogeneity and incomplete state matching (Supplementary Note 2.5).

Data preprocessing

All datasets were processed using the Scanpy Python package[66]. To ensure compatibility between spatial and reference datasets, analyses were restricted to genes shared between the imaging-based spatial transcriptomics dataset and its corresponding reference. All expression matrices were library-size normalized using `sc.pp.normalize_total` and log-transformed using `sc.pp.log1p`.

For spatial datasets, only cells with valid platform-provided annotations were retained. Cells labeled as Unlabeled in the Xenium dataset were excluded, whereas all cells were retained for the MERFISH-based MOp and MBC datasets. Reference scRNA-seq and snRNA-seq datasets were processed according to the quality-control procedures described in the original studies. MOp reference cells were filtered using the author-provided quality-control annotations, whereas no additional filtering was required for the breast cancer and MBC references.

The resulting normalized expression matrices of shared measured genes were used for cell-type annotation, construction of transcriptomic similarity matrices, and FGW-based transcriptome reconstruction. Detailed preprocessing procedures for each dataset are provided in Supplementary Note 3.

Cross-validation on spatially measured genes

To assess transcriptome reconstruction performance, we performed five-fold cross-validation using genes measured in both the spatial and reference datasets. Shared genes were randomly partitioned into five subsets. In each fold, one subset was held out as the test set, whereas the remaining genes were used for model training. Predicted expression values for held-out genes were then compared with their measured spatial expression. This procedure was repeated across all folds to provide an unbiased estimate of reconstruction accuracy.

Benchmarked methods. We benchmarked Emerge against representative reference-based transcriptome reconstruction methods, including stPlus[16], SpaGE[14], Seurat[13], ENGEP[15], Tangram[21], NovoSpaRc[22], SpaIM[12], and SPRITE[25]. These methods represent diverse strategies for integrating single-cell references with spatial transcriptomics data, including shared-gene alignment, latent-space mapping, ensemble learning, and geometry-aware optimal transport approaches. For each method, we used the authors' implementations and recommended hyperparameters while following the corresponding preprocessing procedures to ensure fair comparison. Due to the substantially larger scale of the Xenium breast cancer dataset, NovoSpaRc[22] and SPRITE[25] were not included in the corresponding benchmark analyses because of their high computational requirements on the full dataset.

Benchmark metrics. Performance was evaluated at both gene and cell levels using six complementary metrics: Pearson correlation coefficient (PCC), Spearman correlation coefficient (SCC), structural similarity index (SSIM), Jensen-Shannon divergence (JSD), mean absolute error (MAE), and root mean squared error (RMSE) (definitions provided in Supplementary Note 2.1). Higher PCC, SCC, and SSIM and lower JSD, MAE, and RMSE indicate better reconstruction performance.

Because different evaluation metrics capture complementary aspects of reconstruction performance, no single metric fully characterizes reconstruction quality, and method rankings may vary across evaluation criteria. To provide a unified assessment that is less sensitive to the choice of any individual metric, we adopted a rank-based aggregation strategy following previous benchmarking studies[67]. Rather than directly averaging raw metric values, which operate on different numerical scales and are not directly comparable, we summarized performance using an aggregated accuracy rank (AR) defined as the mean rank across all six metrics:

$$\mathrm{AR} = \frac{1}{6}\left(\mathrm{RANK}_{\mathrm{PCC}} + \mathrm{RANK}_{\mathrm{SCC}} + \mathrm{RANK}_{\mathrm{SSIM}} + \mathrm{RANK}_{\mathrm{JSD}} + \mathrm{RANK}_{\mathrm{MAE}} + \mathrm{RANK}_{\mathrm{RMSE}}\right), \tag{10}$$

where ranking directions were aligned such that higher AR values correspond to better overall reconstruction performance.

Evaluation strategy. Because Emerge performs type-restricted reconstruction and explicitly models microenvironment-dependent variation, we evaluated predictive accuracy at two complementary levels: (1) global accuracy across all shared genes and cell types, and (2) within-cell-type accuracy on environmentally structured markers. This dual evaluation design mirrors the modeling principles of Emerge and allows a direct assessment of its ability to preserve both broad identities and microenvironment-dependent within-type variation.

Global evaluation. Global performance was evaluated using five-fold cross-validation across all shared genes. Reconstruction accuracy was quantified using PCC, SCC, SSIM, JSD, MAE, RMSE, and AR. This evaluation reflects overall transcriptome-scale prediction performance.

Within-cell-type evaluation. To evaluate recovery of microenvironment-associated variation within cell types, we performed type-specific analyses using structured genes. Structured genes were identified exclusively from measured expression data prior to reconstruction and were required to exhibit both substantial intra-type transcriptional variability and spatial organization according to predefined differential-expression and spatial-autocorrelation criteria. Because Emerge is specifically designed to recover microenvironment-associated transcriptional variation, these genes provide a targeted evaluation of the biological signals most directly related to the modeling objective. For each cell type, gene-wise correlations between measured and predicted expression were computed and summarized analogously to the global evaluation.

To assess robustness to gene-set definition, we additionally evaluated highly variable genes (HVGs), non-HVGs, spatially variable genes (SVGs), non-SVGs and curated marker genes using the same evaluation framework. Detailed definitions of all gene sets, selection criteria and evaluation procedures are provided in Supplementary Note 2.2.

Effect of cell-type annotation settings

Because Emerge relies on cell-type labels to define type-constrained alignment, reconstruction performance may be influenced by upstream annotation settings. We therefore systematically evaluated robustness with respect to four annotation-related factors: annotation method, annotation granularity, confidence threshold and annotation mode.

To assess robustness to annotation method, cell-type labels generated by alternative annotation approaches were used as inputs to Emerge and reconstruction performance was evaluated using the same benchmarking framework. To assess robustness to annotation granularity, analyses were performed using both coarse-grained and fine-grained annotations. Coarse-grained annotations correspond to major cell classes, whereas fine-grained annotations represent subtype-level classifications when available. To evaluate the effect of annotation confidence, cells were divided into high-confidence and low-confidence groups according to the maximum consensus assignment probability. The confidence threshold was varied across a range of values to assess its impact on reconstruction performance and alignment behavior. In addition, we compared the default single-cell mode (Emerge_cell) with the cluster-level mode (Emerge_cluster), in which annotation probabilities are aggregated within spatial clusters before consensus assignment.

For all analyses, reconstruction accuracy was evaluated using the same metrics and cross-validation procedures described above. Detailed experimental settings and evaluation procedures are provided in Supplementary Note 2.3.

External validation using orthogonal spatial assays

To assess whether Emerge recovers biologically meaningful spatial patterns for genes not measured by imaging-based assays, we performed external validation using independent spatial modalities rather than cross-validation on panel genes.

For the mouse primary motor cortex (MOp), Emerge was applied using all MERFISH-measured genes to generate transcriptome-scale predictions. Predicted spatial expression patterns were compared with corresponding in situ hybridization (ISH) images from the Allen Brain Atlas[47]. Because the ISH data provide image-based anatomical reference maps rather than directly matched single-cell measurements, this analysis was used as a qualitative validation of reconstructed spatial expression patterns and anatomical localization.

For the human breast cancer dataset, predicted full-transcriptome profiles were validated using an adjacent Visium section that provides genome-wide spatial expression measurements. Following the strategy of Janesick et al.[4], predicted single-cell expression profiles were aggregated into pseudo-Visium spots according to the spatial layout of the matched Visium measurements. For each gene, spot-level expression values from the pseudo-Visium data were compared with the corresponding measured Visium expression values, and Pearson correlation coefficients were computed across spots to quantify agreement between predicted and observed spatial expression patterns. Representative genes were additionally visualized to assess spatial concordance.

Together, these complementary qualitative and quantitative validations assess whether Emerge can recover biologically meaningful spatial expression patterns beyond genes directly measured by imaging-based spatial transcriptomics assays.

Statistics and reproducibility

This study is based on computational analyses of publicly available spatial transcriptomics and single-cell transcriptomics datasets. No new experimental data were generated. Sample sizes were determined by the availability of the corresponding public datasets, and no statistical method was used to predetermine sample size. No data were excluded from the analyses unless explicitly stated in the preprocessing procedures. The experiments were not randomized, and the investigators were not blinded to allocation during experiments and outcome assessment, as this study involved analysis of previously generated public datasets.

Statistical analyses were performed using the methods described in the manuscript and Supplementary Information. Statistical significance was assessed using two-sided paired Wilcoxon signed-rank tests where indicated, and enrichment analyses were performed using hypergeometric tests. All computational analyses were performed using the publicly available Emerge source code and analysis scripts to facilitate reproducibility.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets analyzed in this study are publicly available from the following sources. For the MERFISH MOp dataset, the spatial transcriptomics dataset[41] was obtained from the https://doi.org/10.35077/g.21, and the corresponding sc/snRNA-seq dataset[42] was accessed through the BRAIN Initiative Cell Census Network (https://assets.nemoarchive.org/dat-ch1nqb7). For the Xenium breast cancer dataset[4], the Xenium in situ spatial data, matched scRNA-seq data, and Visium spatial data were all obtained from 10x Genomics at https://www.10xgenomics.com/products/xenium-in-situ/preview-dataset-human-breast. The independent scRNA-seq reference[52] was obtained from https://singlecell.broadinstitute.org/single_cell/study/SCP1039. For MERFISH MBC datasets[58], the spatial datasets and corresponding scRNA-seq (snRNA-seq) datasets of all 12 tissue sections were downloaded from the Single-Cell Portal at https://singlecell.broadinstitute.org/single_cell/study/SCP2702. For validation, in situ hybridization images were obtained from the Allen Mouse Brain Atlas[47]. Any additional requests for information can be directed to and will be fulfilled by the corresponding authors. Source data are provided with this paper.

Code availability

The Emerge software package and source code are available on GitHub (https://github.com/st-yang97/Emerge) and archived on Zenodo (https://doi.org/10.5281/zenodo.21604723)[68]. All scripts and materials required to reproduce the analyses are available at the same repository.

References

1. Schürch, C. M. et al. Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front. *Cell* **182**, 1341–1359.e19 (2020).
2. Baccin, C. et al. Combined single-cell and spatial transcriptomics reveal the molecular, cellular and spatial bone marrow niche organization. *Nat. Cell Biol.* **22**, 38–48 (2020).
3. Moffitt, J. R. et al. Molecular, spatial, and functional single-cell profiling of the hypothalamic preoptic region. *Science* **362**, eaau5324 (2018).
4. Janesick, A. et al. High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis. *Nat. Commun.* **14**, 8353 (2023).
5. He, S. et al. High-plex imaging of RNA and proteins at subcellular resolution in fixed tissue by spatial molecular imaging. *Nat. Biotechnol.* **40**, 1794–1806 (2022).
6. Chen, K. H., Boettiger, A. N., Moffitt, J. R., Wang, S. & Zhuang, X. Spatially resolved, highly multiplexed RNA profiling in single cells. *Science* **348**, aaa6090 (2015).
7. Longo, S. K., Guo, M. G., Ji, A. L. & Khavari, P. A. Integrating single-cell and spatial transcriptomics to elucidate intercellular tissue dynamics. *Nat. Rev. Genet.* **22**, 627–644 (2021).
8. Covert, I. et al. Predictive and robust gene selection for spatial transcriptomics. *Nat. Commun.* **14**, 2091 (2023).
9. Kuemmerle, L. B. et al. Probe set selection for targeted spatial transcriptomics. *Nat. Methods* **21**, 2260–2270 (2024).
10. Tian, L., Chen, F. & Macosko, E. Z. The expanding vistas of spatial transcriptomics. *Nat. Biotechnol.* **41**, 773–782 (2023).
11. Wan, X. et al. Integrating spatial and single-cell transcriptomics data using deep generative models with SpatialScope. *Nat. Commun.* **14**, 7848 (2023).
12. Li, B. et al. SpaIM: single-cell spatial transcriptomics imputation via style transfer. *Nat. Commun.* **16**, 7861 (2025).
13. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902.e21 (2019).
14. Abdelaal, T., Mourragui, S., Mahfouz, A. & Reinders, M. J. T. SpaGE: spatial gene enhancement using scRNA-seq. *Nucleic Acids Res.* **48**, e107 (2020).
15. Yang, S. T. & Zhang, X. F. ENGEP: advancing spatial transcriptomics with accurate unmeasured gene expression prediction. *Genome Biol.* **24**, 293 (2023).
16. Chen, S., Zhang, B., Chen, X., Zhang, X. & Jiang, R. stPlus: a reference-based method for the accurate enhancement of spatial transcriptomics. *Bioinformatics* **37**, i299–i307 (2021).
17. Lopez, R., Regier, J., Cole, M. B., Jordan, M. I. & Yosef, N. Deep generative modeling for single-cell transcriptomics. *Nat. Methods* **15**, 1053–1060 (2018).
18. Cao, K., Gong, Q., Hong, Y. & Wan, L. A unified computational framework for single-cell data integration with optimal transport. *Nat. Commun.* **13**, 7419 (2022).
19. Sun, E. D., Ma, R., Navarro, N. P., Brunet, A. & Zou, J. TISSUE: uncertainty-calibrated prediction of single-cell spatial transcriptomics improves downstream analyses. *Nat. Methods* **21**, 444–454 (2024).

20. Ma, R., Sun, E. D., Donoho, D. & Zou, J. Principled and interpretable alignability testing and integration of single-cell data. *Proc. Natl. Acad. Sci. USA* **121**, e2313719121 (2024).
21. Biancalani, T. et al. Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram. *Nat. Methods* **18**, 1352–1362 (2021).
22. Nitzan, M., Karaiskos, N., Friedman, N. & Rajewsky, N. Gene expression cartography. *Nature* **576**, 132–137 (2019).
23. Cang, Z. & Nie, Q. Inferring spatial and signaling relationships between cells from single cell transcriptomic data. *Nat. Commun.* **11**, 2084 (2020).
24. Rahimi, A., Vale-Silva, L. A., Fälth Savitski, M., Tanevski, J. A. & Saez-Rodriguez, J. DOT: a flexible multi-objective optimization framework for transferring features across single-cell and spatial omics. *Nat. Commun.* **15**, 4994 (2024).
25. Sun, E. D., Ma, R. & Zou, J. SPRITE: improving spatial gene expression imputation with gene and cell networks. *Bioinformatics* **40**, i521–i528 (2024).
26. Armingol, E., Officer, A., Harismendy, O. & Lewis, N. E. Deciphering cell-cell interactions and communication from gene expression. *Nat. Rev. Genet.* **22**, 71–88 (2021).
27. Damond, N. et al. A map of human type 1 diabetes progression by imaging mass cytometry. *Cell Metab.* **29**, 755–768.e5 (2019).
28. Ji, A. L. et al. Multimodal analysis of composition and spatial architecture in human squamous cell carcinoma. *Cell* **182**, 497–514.e22 (2020).
29. Zhang, W. et al. Non-invasive profiling of the tumour microenvironment with spatial ecotypes. *Nature* **654**, 1076–1087 (2026).
30. He, H.-F. et al. Unlocking single-cell level and continuous whole-slide insights in spatial transcriptomics with PanoSpace. *Nat. Comput. Sci.* **6**, 358–371 (2026).
31. Tu, J. J., Yan, H., Zhang, X.-F. & Lin, Z. Precise gene expression deconvolution in spatial transcriptomics with STged. *Nucleic Acids Res.* **53**, gkaf087 (2025).
32. Wang, M. G., Zhang, X. F. & Chen, L. Dual decoding of cell types and gene expression in spatial transcriptomics with PANDA. *Nucleic Acids Res.* **52**, 12173–12190 (2024).
33. Birk, S. et al. Quantitative characterization of cell niches in spatially resolved omics data. *Nat. Genet.* **57**, 897–909 (2025).
34. Varrone, M., Tavernari, D., Santamaria-Martínez, A., Walsh, L. A. & Ciriello, G. CellCharter reveals spatial cell niches associated with tissue remodeling and cell plasticity. *Nat. Genet.* **56**, 74–84 (2024).
35. Kim, H. et al. CellNeighborEX: deciphering neighbor-dependent gene expression from spatial transcriptomics data. *Mol. Syst. Biol.* **19**, e11670 (2023).
36. Mason, K. et al. Niche-DE: niche-differential gene expression analysis in spatial transcriptomics data identifies context-dependent cell-cell interactions. *Genome Biol.* **25**, 14 (2024).
37. Haviv, D. et al. The covariance environment defines cellular niches for spatial inference. *Nat. Biotechnol.* **43**, 269–280 (2025).
38. Tejada-Lapuerta, A. et al. Nicheformer: a foundation model for single-cell and spatial omics. *Nat. Methods* **22**, 2525–2538 (2025).
39. Liu, Y. et al. Conserved spatial subtypes and cellular neighborhoods of cancer-associated fibroblasts revealed by single-cell spatial multi-omics. *Cancer Cell* **43**, 905–924.e6 (2025).
40. Titouan, V., Courty, N., Tavenard, R., Chapel, L. & Flamary, R. Optimal transport for structured data with application on graphs. In *International Conference on Machine Learning* 97 (eds Chaudhuri, K. & Salakhutdinov, R.) 6275–6284 (PMLR, 2019).
41. Zhang, M. et al. Spatially resolved cell atlas of the mouse primary motor cortex by MERFISH. *Nature* **598**, 137–143 (2021).
42. Yao, Z. et al. A transcriptomic and epigenomic cell atlas of the mouse primary motor cortex. *Nature* **598**, 103–110 (2021).
43. Kiselev, V. Y., Yiu, A. & Hemberg, M. Scmap: projection of single-cell RNA-seq data across data sets. *Nat. Methods* **15**, 359–362 (2018).
44. Lin, Y. et al. ScClassify: sample size estimation and multiscale classification of cells using single and multiple reference. *Mol. Syst. Biol.* **16**, e9389 (2020).
45. Singhal, V. et al. BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nat. Genet.* **56**, 431–441 (2024).
46. Liu, Q. et al. Microenvironment-aware spatial modeling for accurate inference of cell identity. *Nucleic Acids Res.* **54**, gkaf1477 (2026).
47. Lein, E. S. et al. Genome-wide atlas of gene expression in the adult mouse brain. *Nature* **445**, 168–176 (2007).
48. Khakh, B. S. & Sofroniew, M. V. Diversity of astrocyte functions and phenotypes in neural circuits. *Nat. Neurosci.* **18**, 942–952 (2015).
49. Batiuk, M. Y. et al. Identification of region-specific astrocyte subtypes at single cell resolution. *Nat. Commun.* **11**, 1220 (2020).
50. Mathys, H. et al. Single-cell transcriptomic analysis of Alzheimer's disease. *Nature* **570**, 332–337 (2019).
51. Hasel, P. et al. Neurons and neuronal activity control gene expression in astrocytes to regulate their development and metabolism. *Nat. Commun.* **8**, 15132 (2017).
52. Wu, S. Z. et al. A single-cell and spatially resolved atlas of human breast cancers. *Nat. Genet.* **53**, 1334–1347 (2021).
53. Zhou, C., Gao, Y., Ding, P., Wu, T. & Ji, G. The role of CXCL family members in different diseases. *Cell Death Discov.* **9**, 212 (2023).
54. Tokunaga, R. et al. CXCL9, CXCL10, CXCL11/CXCR3 axis for immune activation—a target for novel cancer therapy. *Cancer Treat. Rev.* **63**, 40–47 (2018).
55. Pelka, K. et al. Spatially organized multicellular immune hubs in human colorectal cancer. *Cell* **184**, 4734–4752.e20 (2021).
56. Binnewies, M. et al. Understanding the tumor immune microenvironment (TIME) for effective therapy. *Nat. Med.* **24**, 541–550 (2018).
57. Costa, A. et al. Fibroblast heterogeneity and immunosuppressive environment in human breast cancer. *Cancer Cell* **33**, 463–479.e10 (2018).
58. Klughammer, J. et al. A multi-modal single-cell and spatial expression map of metastatic breast cancer biopsies across clinicopathological features. *Nat. Med.* **30**, 3236–3249 (2024).
59. Deng, S. et al. MYL9 expressed in cancer-associated fibroblasts regulate the immune microenvironment of colorectal cancer and promotes tumor progression in an autocrine manner. *J. Exp. Clin. Cancer Res.* **42**, 294 (2023).
60. Yu, Z. H. et al. Integrating spatial and single-cell transcriptomes reveals the role of COL1A2^{+} MMP1$^{+/-}$ cancer-associated fibroblasts in ER-positive breast cancer. *Cancer Cell Int.* **25**, 82 (2025).
61. Xue, T. et al. Prognostic significance of CD163^{+} tumor-associated macrophages in colorectal cancer. *World J. Surg. Oncol.* **19**, 186 (2021).
62. Kalluri, R. The biology and function of fibroblasts in cancer. *Nat. Rev. Cancer* **16**, 582–598 (2016).
63. Jin, S. et al. Inference and analysis of cell-cell communication using CellChat. *Nat. Commun.* **12**, 1088 (2021).
64. Xiong, Y. X., Wang, M. G., Chen, L. & Zhang, X. F. Cell-type annotation with accurate unseen cell-type identification using multiple references. *PLoS Comput. Biol.* **19**, e1011261 (2023).
65. Nguyen, N. D. et al. scDOT: optimal transport for mapping senescent cells in spatial transcriptomics. *Genome Biol.* **25**, 288 (2024).
66. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 15 (2018).
67. Li, B. et al. Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution. *Nat. Methods* **19**, 662–670 (2022).
68. Yang, S. T. et al. Emerge: microenvironment-aware transcriptome reconstruction in spatial transcriptomics. *Zenodo* https://doi.org/10.5281/zenodo.21604723 (2026).

Acknowledgements
We thank the authors and contributors of the publicly available spatial transcriptomics and single-cell RNA sequencing datasets used in this study. We also thank the members of our research groups for valuable discussions and suggestions.

Author contributions
This study was conceived and led by X.F.Z. and L.C. S.T.Y. developed and performed data analyses. S.T.Y. and X.F.Z. wrote the manuscript. P.P., H.F.H., M.G.W., and B.H.S. assisted with data analyses and manuscript writing. X.F.Z. and L.C. supervised the entire project. All authors read and approved the final version.

Funding
This work was supported by the National Natural Science Foundation of China [12271198, 11871026, 62377019, 12131020, 31930022, T2350003, T2341007, 42450084, 42450135, 42450192, 12326614 and 12501724], Academy of Frontier Interdisciplinary Research, Central China Normal University [JC2026PT-002], National Key R&D Program of China (2022YFA1004800, 2025YFF1207900), Zhejiang Province Vanguard Goose-Leading Initiative (2025C01114), Science and Technology Commission of Shanghai Municipality [23JS1401300], and JST Moonshot R&D [JPMJMS2021].

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-77349-8.

Correspondence and requests for materials should be addressed to Xiao-Fei Zhang or Luonan Chen.

Peer review information *Nature Communications* thanks the anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026